Inter-Sensor High-Resolution and Multi-Temporal Image Fusion for Unsupervised Domain Adaptation in Remote Sensing

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***Abstract*— Motivated by the increasing demand for robust segmentation in unlabeled remote sensing data, we propose domain adaptation multimodal and multi-temporal transformer (DAM-Former), a novel unsupervised domain adaptation (UDA) model that fuses high-resolution (HR) multimodal imagery with multi-temporal multispectral data. Current UDA approaches in remote sensing rarely exploit the complementary strengths of spatial and temporal features. To address this gap, our framework integrates two interconnected branches: a transformer-based network for HR multimodal data and a lightweight convolutional network with temporal attention for multi-temporal imagery. To improve segmentation accuracy and lower noise, the extracted features are robustly combined through a deep temporal fusion (DTF) module and a new mixed loss (ML) with an ensemble pseudo-label (EP) strategy. Extensive experiments and an ablation study on the FLAIR-2 dataset demonstrate that DAM-Former outperforms state-of-the-art methods, marking the first in-depth study of temporal information fusion in UDA segmentation for remote sensing data. Code available at https://github.com/ibanezfd/DAM_Former.**

***Index Terms*— Data fusion, image segmentation, multimodal, remote sensing, unsupervised domain adaptation (UDA).**

I. Introduction

THE ability to assign a class or label to each pixel of an image, also known as image segmentation, has become one of the most common computer vision tasks [1], [2]. A wide variety of fields require identifying the different areas or objects composing an image, such as autonomous driving [3], [4], medical imaging [5], [6], or remote sensing [7], [8], [9]. In the particular case of remote sensing, the use of image segmentation techniques provides solutions for a number of applications, including land cover [10], disaster response [11], [12], agricultural land use [13], [14], and environmental management [15], to name a few. However, remote sensing data processing faces specific challenges not found in other kinds of images. First, remote sensing images are usually provided by multispectral or hyperspectral sensors, which acquire non-conventional data. At the same time, remote sensing images contain a large amount of information per pixel compared to other images, and therefore, pixel or subpixel accuracy is crucial [16]. Nevertheless, one of the most significant challenges in remote sensing has been the lack of accessible data to train models. It was not until recent years, that there has been an increase in the number of space and aerial missions providing public remote sensing data. In [17], the training of larger deep learning models has been made possible, leading to notable improvements in the performance of semantic segmentation models across various computer vision domains.

Nowadays, thanks to the Sentinel missions [18] of the Copernicus Program, the Landsat missions [19], or the Google Earth Engine [20] among others, there is a large amount of publicly available remote sensing data [21]. However, the vast majority of these available images are unlabeled [22], creating a new challenge for remote sensing. The process of labeling remote sensing data is a tedious task that consumes a high number of resources and time [23]. Several approaches to confront the lack of labeled data are currently being investigated, such as using unsupervised [24] or semi-supervised learning strategies [25]. Semi-supervised learning strategies commonly achieve higher performances than unsupervised methods [26]. Still, they require at least some labeled data from the images to segment. In contrast, unsupervised methods, which do not require labels of the target data to segment, use different techniques to inherit previous knowledge and then adapt the models to the target data. One of these techniques is unsupervised domain adaptation (UDA) [27].

The objective of UDA is to transfer the learned representations from a first set of images whose labels are known (the source domain) to a second set of images whose labels are unknown (the target domain). Those two domains could contain images of different times, locations, instruments, or even use simulated data sources. Using very different source and target domains can ease the task of finding a labeled

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source dataset. However, when the differences between the two domains increase, the transfer of information between the target and source representations becomes increasingly difficult. This problem is also known as the domain gap [28], and many works in UDA focus on reducing this gap between the domains.

In the area of remote sensing, several UDA methods have been proposed [29]. Bruzzone and Marconcini [30] researched the use of UDA with support vector machines (SVMs) for Landsat-5 imagery. Schmitter et al. [31] also adopted UDA strategies with SVM to detect stress levels in vegetation from hyperspectral images. In image segmentation tasks, it was not until the emergence of neural networks in more recent times that UDA techniques increased their presence in remote sensing, particularly with adversarial generative networks. Liu and Su [32] used adversarial training with a DeepLab V3+ neural network to reduce the domain gap between remote sensing data, adapting the deep features of the network into curve images and training a curve feature extractor to reduce the discrepancy between domains. A different approach was performed in [33], where two adversarial networks were co-trained using a category-sensitive domain adaptation for land cover.

In recent times, after the transformer architecture [34] achieved great success in natural language processing and computer vision tasks [35], different approaches have been studied for it in remote sensing scenarios, including UDA tasks. Li et al. [36] proposed a self-trained transformer based on the DAFormer structure [37], which includes stabilizing class balancing and thresholding of the pseudo-labels. Ma et al. [38] defined a different framework for very high-resolution (HR) remote sensing images with a generative network based on a transformer architecture and a multidiscrimination module.

One of the advantages of using remote sensing data to identify pixel information is the wide variety of current instruments for obtaining images. However, the use of different data sources in a multimodal fashion to improve segmentation accuracy through UDA processes has not been explored until recently. For instance, Zou et al. [39] proposed a multimodal generator composed of different networks to extract features from hyperspectral, multispectral, and radar images following an adversarial strategy. Zhou et al. [40] fused digital surface model (DSM) data with remote sensing images at multiple levels with attention-aware blocks. Exploiting the DSM data along with the remote sensing images improved the classification of objects with different heights and shades.

Another surprisingly barely explored area of UDA in remote sensing is the use of time series as input data. The current availability of public remote sensing data on platforms such as the Copernicus Open Access Hub provides temporal images of different Sentinel missions, yet temporal information is still not commonly used for UDA in image segmentation tasks. To our knowledge, only a couple of examples can be found in the literature where temporal information was employed for UDA. One of these works includes the use of UDA techniques with multi-temporal remote sensing images in order to segment invasive wetland plant species [41]. In this case, an intermediate-domain prototype network is used to improve cross-temporal class interpretability. Capliez et al. also used a time series of remote sensing images, combining adversarial and self-training techniques with an encoder that employs of 1-D convolutions in the temporal dimension to extract deep temporal features.

In this context, where sparse work has been done on multimodal input data, and on using time series for UDA in remote sensing image segmentation, the FLAIR-2 dataset [42] emerged. The FLAIR-2 dataset contains remote sensing images for the semantic segmentation of different land cover classes in two modalities. The first modality contains HR aerial images with optical, near-infrared (NIR), and elevation information. The second modality contains a time series of multispectral Sentinel-2 images. The dataset covers most departments of the entire inland territory of France. As the fourth largest country in Europe, covering 543 940 km^2, France includes different scenes, such as coastal, urban, agricultural, rural, or mountainous areas along the country. For these reasons, the FLAIR-2 dataset brings new opportunities to research data fusion techniques for multimodal, multi-temporal remote sensing data. Particularly to transfer segmentation learning between different areas of the country, which have been mapped and classified, to the rest of the French territory. However, the use of multimodal and multi-temporal data has not yet been explored to improve segmentation accuracy in unlabeled remote sensing images through UDA.

In this work, we propose domain adaptation multimodal and multi-temporal transformer (DAM-Former), a baseline framework for multimodal domain adaptation image segmentation tasks, particularly for remote sensing data. Despite the increasing collection of multimodal remote sensing datasets containing multi-temporal series of images, previous research has never studied in-depth the fusion of spatial and temporal modalities in UDA. To address this gap, DAM-Former features two interconnected branches: a U-Net-based transformer encoder for HR multimodal data and a lightweight convolutional network with temporal attention for multi-temporal and multispectral data. While the HR main branch is able to extract precise spatial information of different modalities, the temporal branch identifies class-unique patterns in the temporal dimension.

Both branches are combined at their deepest representation via a deep temporal fusion (DTF) module, enabling the complementary strengths of precise spatial texture and shifting temporal patterns to be enhanced together, without losing precise spatial information due to saturation of redundant features. Moreover, a novel mixed loss (ML) function further fuses the two branches' information, stabilizing the teacher–student self-training and reducing the noise using a new ensemble pseudo-label (EP) framework. Instead of treating outputs from different modalities separately, our approach generates balanced EPs by combining the predictions from both branches. This fusion creates a more robust labeling that mitigates noise and compensates for the weaknesses of any single branch. This enables DAM-Former to provide high-accuracy segmentation predictions robust against domain shifts through the deep integration of temporal information.

Unlike conventional UDA methods that only focus on spatial features, DAM-Former exploits the temporal patterns

in multispectral time series, an aspect that has not been thoroughly explored. In this article, we perform for the first time an extensive analysis of the effects of multimodal and multi-temporal fusion on semantic segmentation UDA. To this end, and to verify the benefits of our method, we perform an UDA segmentation between North and South domains in the FLAIR-2 dataset. We assess how multi-temporal information enhances models to adapt between the wet and cold Atlantic north part of the country, predominated by intensive production of cereals, grains, and polycultures, and the dryer and hotter Mediterranean south part, containing vineyards, orchards, and forestry. In the experiments, we observe how DAM-Former with it's new fusion strategies is able to outperform state-of-the-art similar methods, even when they include temporal information. We also evaluate our model with an ablation study for the different proposed modules, Oracle performance, complexity, and fusion approaches. The main contributions of this work can be summarized as follows.

1) We propose DAM-Former, an UDA-based segmentation method for multimodal and multi-temporal remote sensing data fusion. Our model combines transformer-based and convolutional architectures, using novel fusion techniques including an EP strategy, an ML function, and a DTF module. Fusing information from different modalities at different levels enables stable predictions, mitigates noise, and enforces consistency between branches during training. Exploiting multimodal data improves differentiation between similar classes, such as different vegetation or materials, while maintaining HR segmentation.
2) We explore for the first time the use and fusion of multi-temporal data with other modalities for remote sensing UDA tasks. We analyze the benefits of DAM-Former for domain adaptation under different environmental conditions. Specifically, we focus on a North-South domain adaptation and vice versa analysis of the French territory. Our experiments, including ablation studies, demonstrate significant improvements in segmentation accuracy and better generalization over state-of-the-art methods for both land cover and rare class tasks.

The remaining parts of this article are organized into five main sections. In Section II, a review of the literature is provided, exposing the main limitations of related works in the remote sensing field. In Section III, the technical details of the UDA strategy and the multimodal segmentation architecture of the DAM-Former are described. Section IV contains a description of the FLAIR-2 dataset and the domains selected for this work. Section V contains the experiments performed to study the performance of the proposed model, including an ablation study, extensive class classification results analysis, and other model analysis experiments. Finally, Section VI presents the final thoughts of this work, summarizing the main contributions, limitations, and future challenges.

II. Related Work

A. Semantic Segmentation

The emergence of convolutional neural networks (CNNs) completely changed the field of computer vision, including semantic segmentation [43]. Since then, various advances have made CNN models more accurate or stable, such as different architectures [5], [44], the incorporation of contextual information [45], or attention mechanisms [46], [47]. In the field of remote sensing, CNNs also had a great impact on image segmentation methods, moving from traditional machine learning techniques [9] to CNN-based models [8], [16], including different architectures [48], [49], [50]. However, due to the complexity of remote sensing data, specific adaptations have always been required, such as multispectral interband fusion techniques [51], generating and pretraining with multispectral synthetic images [52], or fusing color and DSM information [53]. Recently, the transformer architecture [34] breakthrough in computer vision [35], [54] has made it possible to better highlight relevant deep features and find mid- and long-range dependencies [55] between features compared to CNNs. In remote sensing, the transformer architecture has also been adopted [56] and adapted, for example, to hyperspectral data [57], [58]. However, most of these methods assume a large amount of labeled data to train these models, which is not always available. In these cases, transfer learning techniques are used to extract information from labeled data and use it with unlabeled or sparsely labeled data.

B. Unsupervised Domain Adaptation

Among the various transfer learning techniques, when the real labels are nonexistent (target domain), but a similar or related series of labeled images are available (source domain) and can be used to extrapolate information, the most common approach is UDA. Several different schemes have been explored to solve UDA tasks in computer vision [27] and specifically in remote sensing [29]. UDA methods are usually divided into four categories according to how the models are trained and the information transferred. First, generative training methods generally pursue generating synthetic data from the source domain, conserving the labels with images similar to the target domain. Then, they use these images for the final training [59], [60], relying greatly on the quality of the samples generated for the training. The second group includes methods where adversarial training aims to reduce the discrepancies between domains [61], [62], which usually requires precise tuning of hyperparameters and proves challenging to adapt to different types of data. The third group consists of self-trained models. These models use pseudo-labels in their training, generated from highly confident predictions of the unlabeled data [63], [64], [37]. Most self-training methods face the issue of having a bias for easy and more predictable classes from imbalanced datasets. However, self-training methods have shown better stability and results than generative adversarial training strategies [65]. The last group contains mixed models, such as generative adversarial networks (GANs) [66], [67] and other combinations [68], [69], which tend to encounter convergence difficulties due to the complexity of their schemes. Moreover, most UDA methods in computer vision are designed only for RGB images and are unable to exploit images of different modalities. Even though

there are multimodal computer vision UDA methods, they are designed for image and text [70] or image and audio [71] multimodal information, which greatly differs from remote sensing multimodal data of aerial, multispectral, hyperspectral, synthetic aperture radar (SAR), light detection and ranging (LiDAR), and other common modalities.

C. Multimodal Image Processing

Due to the vast amount of available remote sensing data and the variety of instruments used to collect this data, deep learning multimodal fusion techniques have been widely explored in recent times [72]. Some of the most common tasks include image registration [73], change detection [74], time series enhancement [75], or segmentation [76]. In the past, handcrafted features were designed to deal with fusion tasks. However, with CNN-based deep learning models, different general methods have been adopted for multimodal data. For example, some networks have been designed to extract multimodal features by concatenating the different modalities in the input [77]. More complex multimodal architectures, including siamese architectures [74], [78], contain two identical networks with shared weights. Two-branched networks [75], [79], for intricate fusion tasks with images of different sizes, are also a popular approach.

D. Key Differences With Related Work

In the previously revised literature, we find a shortage of research in multimodal and multi-temporal data fusion methods in the area of domain adaptation. For that reason, we propose DAM-Former, a new framework to analyze the possible benefits of multimodal and multi-temporal data in UDA segmentation tasks, particularly in remote sensing image data. Our model's architecture is divided into two interconnected branches. For the main branch, we use an HR multimodal network, which is responsible for extracting and fusing features from different modalities of equal high resolution. Based on the results of previous works [37], [54], [57], [58], we use a transformer-based encoder, which is able to extract precise interconnected features and textures along an extensive, HR image.

However, in contrast to the lightweight dense layer decoder adopted in previous works, we adopt a U-Net architecture with a light convolutional decoder. This structure can generate more stable segmentation predictions while maintaining the information through skip connections. The secondary branch is a temporal low-resolution (LR) network able to process a time series of multispectral remote sensing images based on [47]. The temporal convolutional network is faster than the main branch and utilizes temporal attention to identify patterns in this dimension. In order to fuse the information of both branches, the HR multimodal features and the temporal-aware features are connected in the deeper representations of the HR network through a DTF module. The combined use of these two branches allows the DAM-Former to classify pixels for land cover tasks precisely. In comparison to unimodal models, our method is able to differentiate similar classes by leveraging specific multimodal information, such as the height of structures, vegetation, or material signatures in multispectral bands, or vegetation-specific processes only perceived over time.

The entire architecture follows a teacher–student self-training strategy, outperforming generative and adversarial strategies in terms of accuracy and stability [65]. Compared to previous multimodal models, where two branches were concatenated [77] or used shared weights [74], we propose a new ML to further exploit multimodal information. The ML guides our model to generate multimodal EPs, fusing information from both branches, reducing the noise, and allowing for a more stable prediction. We include in our training strategy domain adaptation via cross-domain mixed sampling (DACS) [80], mixing images from the two domains along with the corresponding labels and pseudo-labels and using them as augmented data. We also adopt rare class sampling (RCS), which has been widely adopted in successful computer vision UDA methods such as DAFormer [37] and in remote sensing in [36].

III. Methodology

A. Method Overview

This work introduces DAM-Former, a transformer-based method for UDA in multimodal segmentation. Specifically, DAM-Former is a new baseline model designed to extract and fuse information from several modalities, including LR multispectral time series and HR RGB, NIR, and elevation information. For this purpose, our method follows a self-training strategy (see Section III-B), allowing for stable training. The model architecture (see Section III-C) is divided into three main blocks. The first two blocks define the two very different branches of the model: an HR multimodal network (see Section III-C1), with a transformer-based architecture for the encoder, able to identify mid- and long-range dependencies and patterns across extensive HR images; and a multi-temporal multispectral lighter network (see Section III-C2), based on convolutions with temporal attention, for lower resolution time series of multispectral images. Once the model is able to exploit the different remote sensing modalities, the information is fused into the deeper features of the HR branch (see Section III-C3).

B. Training Strategy

In UDA for image segmentation, two datasets from different domains are considered. First, a source domain dataset $\mathcal{D}_S = (x_S, y_S)$, which contains a set of images x_S and their corresponding labels y_S. Second, a different target domain dataset $\mathcal{D}_T = (x_T)$, where only the images x_T are accessible during training. The final objective of UDA is to use the information from the source domain, $\mathcal{D}_S$, to improve the precision of segmenting the images from the target domain, $\mathcal{D}_T$. In this work, we focus on a multimodal remote sensing image segmentation task. In this case, both datasets, $\mathcal{D}_S$ and $\mathcal{D}_T$, contain different modalities of images. For instance, two main data sources are used: aerial HR images (x_{S_A}), which can contain different modalities, and satellite multi-temporal multimodal images of lower resolution (x_{S_B}). Therefore, we define

our source domain as $\mathcal{D}_S = (x_{S_A}, x_{S_B}, y_S)$. In this case, the two image inputs are $x_{S_A} \in \mathbb{R}^{H_A \times W_A \times M_A}$ and $x_{S_B} \in \mathbb{R}^{H_B \times W_B \times M_B \times T}$, where $H_A \times W_A$ and $H_B \times W_B$ define the size of the spatial dimensions of each data source, M_A contains the different channels and modalities for the aerial image, M_B contains the different spectral dimensions of the satellite image, and T is the temporal number of samples. The labels of the source domain are defined as $y_S \in \mathbb{R}^{H_A \times W_A \times C}$, with the same spatial size as the HR image, and C is the number of classes to segment. The target domain $\mathcal{D}_T = (x_{T_A}, x_{T_B})$ contains the same type of images from a different location, $x_{T_A} \in \mathbb{R}^{H_A \times W_A \times M_A}$ and $x_{T_B} \in \mathbb{R}^{H_B \times W_B \times M_B \times T}$; however, the ground truth is unknown.

In order to apply UDA, a self-training strategy with a teacher–student framework was chosen. In this strategy, two identical networks are used: a student network g_θ with parameters θ and a teacher network h_ϕ with parameters ϕ. During training, the teacher network h_ϕ has the objective of generating pseudo-labels for the target domain $\mathcal{D}_T$. In this case, no augmentation is performed on the target domain data, increasing the efficiency of the training. We propose the use of EPs p_{T_e}, taking advantage of the multimodal information. The EPs are obtained as follows:

$$\left(p_{T_e}^{(i,j)}\right)_c = \left[c = \arg\max_{c'} \left(\frac{\beta \cdot p_{T_A}^{(i,j)} + \gamma \cdot p_{T_B}^{(i,j)}}{\beta + \gamma} \right)_{c'} \right] \tag{1}$$

$$(p_{T_A}, p_{T_B}) = h_\phi(x_{T_A}, x_{T_B}) \tag{2}$$

where $(p_{T_e}^{(i,j)})_c$ is the cth scalar of a one-hot vector defining the EP corresponding to the pixel located at (i, j) in the complete pseudo-labeled image p_{T_e}. Each pixel pseudo-label is generated if the condition inside the Iverson bracket $[\cdot]$ is satisfied. To generate the EPs, the teacher network h_ϕ generates a main prediction $p_{T_A} \in \mathbb{R}^C$ fusing the complete multimodal information and an auxiliary prediction $p_{T_B} \in \mathbb{R}^C$, obtained only from the temporal information, as shown in (2). The normalized class probabilities from both predictions in each pixel are combined to obtain the final EPs. Inspired by ensemble methods, where the performance of the predictions is improved by combining different models, the multimodal ensemble of predictions for the pseudo-labels reduces the noise and improves the stability of the pseudo-labels. The terms β and γ are used to regularize the importance of each term.

However, the teacher network h_ϕ is not backpropagated after generating the pseudo-labels. The teacher network's parameters ϕ are updated online from the student exponential moving average (EMA) of the student network's parameters θ

$$\phi_{t+1} \leftarrow \alpha \phi_t + (1 - \alpha)\theta_t \tag{3}$$

where t is the training step, and $\alpha \in [0, 1]$ is the teacher momentum, a hyperparameter indicating the importance of the current state of the teacher network's parameters h_{ϕ_t}.

Meanwhile, the student network g_θ is trained with a mix of labeled data from the source domain $\mathcal{D}_S$ and generated pseudo-labels for the target domain $\mathcal{D}_T$ data. The training set for the student network is augmented, following DACS [80] with a cross-domain mixing of source and target domain data, helping to reduce the domain gap and the class conflation problem. The data is further augmented using color jitter, Gaussian Blur, and ClassMix [81]. To balance the data, RCS [37] is also applied to the source domain, increasing the sampling of images with the less common classes during the training process. For the source domain data $\mathcal{D}_S$, the student network g_θ is trained in a supervised way, using a categorical cross-entropy loss. Specifically, we propose an ML, with a term for the HR network and another for the temporal network, which are optimized together, stabilizing the training of both networks. The generated predictions from the student network are defined in (4), the cross entropy loss of each network in (5) and (6), and the final ML for the source domain data is defined in (7)

$$(p_{S_A}, p_{S_B}) = g_\theta(x_{S_A}, x_{S_B}) \tag{4}$$

$$\mathcal{L}_{S_A}(p_{S_A}, y_S) = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{C}\left(y_S^{(i,j)}\right)_c \cdot \log\left(p_{S_A}^{(i,j)}\right)_c \tag{5}$$

$$\mathcal{L}_{S_B}(p_{S_B}, y_S) = -\sum_{i=1}^{H}\sum_{j=1}^{W}\sum_{c=1}^{C}\left(y_S^{(i,j)}\right)_c \cdot \log\left(p_{S_B}^{(i,j)}\right)_c \tag{6}$$

$$\mathcal{L}_S(\mathcal{L}_{S_A}, \mathcal{L}_{S_B}) = \mathcal{L}_{S_A} + \mathcal{L}_{S_B}. \tag{7}$$

In contrast, for the target domain data $\mathcal{D}_T$, a quality function is used to evaluate the confidence of the EPs. For this purpose, we consider the number of pixels surpassing a threshold of the maximum softmax of the combined probabilities, as shown in the following equation:

$$q_T = \frac{\sum_{i=1}^{H}\sum_{j=1}^{W}\left[\max_{c'}\left(\frac{\beta \cdot p_{T_A}^{(i,j)} + \gamma \cdot p_{T_B}^{(i,j)}}{\beta+\gamma}\right)_{c'} > \tau\right]}{H \cdot W} \tag{8}$$

where q_T is the combined quality factor for each EP, and τ is the minimum confidence threshold to exceed in order to consider the pseudo-label. The combined quality factor q_T and the EPs p_T are combined to train the student network $g(\theta)$ in the target domain $\mathcal{D}_T$ with a cross-entropy scheme, as shown in (9), as shown at the bottom of the page 7. The complete training strategy is summarized in Fig. 1. We also include a pseudocode of the training process in Algorithm 1 to outline the key steps of the algorithm. During the inference process, only the teacher-trained network would be used to generate the segmentation maps.

C. Model Architecture

The complete model architecture can be divided into two different networks for each data stream. In Section III-C1, the main network for the HR multimodal aerial data is defined. This network follows an encoder–decoder scheme with transformer-based encoders and a lighter convolutional decoder. The details of the temporal network are defined in Section III-C2. We follow the U-TAE architecture first proposed in [47], an efficient and light convolutional network with temporal attention. In Section III-C3, the DTF module between both branches, which directly interconnects them, is defined. A general diagram of the complete architecture can be seen in Fig. 2.

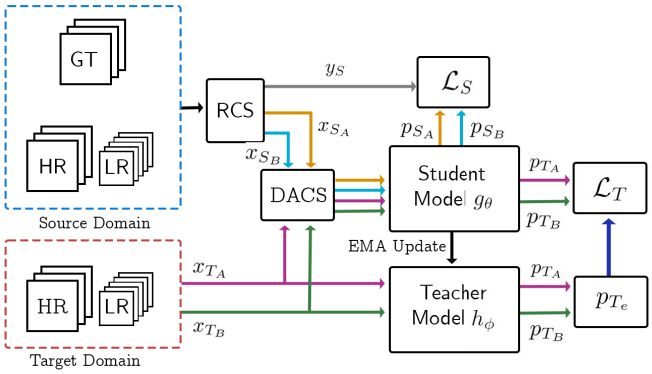


Fig. 1. Training strategy of the DAM-Former. Including the RCS and augmentation (DACS) processes for the input data, the generation of the EPs p_{T_e} and the data used in the ML for the target data $\mathcal{L}_T$ and for the source data $\mathcal{L}_S$.

Algorithm 1 DAM-Former Training Algorithm

Require: Source Domain dataset $\mathcal{D}_S = (x_{S_A}, x_{S_B}, y_S)$, Target Domain dataset $\mathcal{D}_T = (x_{T_A}, x_{T_B})$
Require: Pretrained student network g_θ, Teacher network h_ϕ (initialized as $h_\phi = g_\theta$)
Require: Total iterations N, Learning rate η
Require: Pseudo-label β, Pseudo-label α, Teacher momentum α, Confidence threshold τ

Initialize student network g_θ with ImageNet-pretrained weights
Set teacher network $h_\phi \leftarrow g_\theta$
for $t = 1$ to N **do**
  ▷ *Learning Rate warm-up*
  ▷ *Rare Class Sampling (RCS)*
  ▷ *DACS*
  ▷ *Process Source Batch*
  $\mathcal{L}_{S(batch,a)} = 0$
  for each $(x_{S_A}, x_{S_B}, y_S) \in \mathcal{D}_{S(batch,a)}$ **do**
    Compute prediction: $p_{S_A}, p_{S_B} = g_\theta(x_{S_A}, x_{S_B})$
    Compute source loss (eq. (7)):
    $\mathcal{L}_{S(batch,a)} = \mathcal{L}_{S(batch,a)} + \mathcal{L}_S(p_{S_A}, p_{S_B}, y_S)$
  end for
  ▷ *Process Target Batch (Self-Training)*
  $\mathcal{L}_{T(batch)} = 0$
  for each $(x_{T_A}, x_{T_B}) \in \mathcal{D}_{T(batch)}$ **do**
    Compute teacher prediction:
    $p_{T_A}, p_{T_B} = h_\phi(x_{T_A}, x_{T_B})$
    Generate ensemble pseudo-labels (eq. (1)):
    $p_{T_e}(p_{T_A}, p_{T_B}, \beta, \gamma)$
    Compute confidence score(eq. (8)):
    $q_T(p_{T_A}, p_{T_B}, \beta, \gamma, \tau)$
    Apply DACS augmentation to target dataset:
    $\mathcal{D}_{T(batch,a)} = DACS(\mathcal{D}_{T(batch)})$
    Compute augmented student output:
    $p_{T_{A(a)}}, p_{T_{B(a)}} = g_\theta(x_{T_{A(a)}}, x_{T_{B(a)}})$
    Compute target loss (eq. (9)):
    $\mathcal{L}_{T(batch)} = \mathcal{L}_{T(batch)} + \mathcal{L}_T(p_{T_{A(a)}}, p_{T_{B(a)}}, p_{T_e})$
  end for
  ▷ *Total Loss and Student Model Update*
  $\mathcal{L}_{total} = \mathcal{L}_S + \mathcal{L}_T$
  Update $g(\theta)$ using learning rate lr
  ▷ *Update Teacher Network (EMA)* (eq. (3)):
  $\phi \leftarrow \alpha\phi + (1-\alpha)\theta$
end for
return trained teacher network h_ϕ

1) HR Multimodal Network: The main network is in charge of obtaining multimodal features and generating HR segmentation images. This network has been designed with an encoder–decoder scheme similar to a U-Net [5]. In this network, a series of encoders reduces and extracts deeper features from the images. Then, a decoder recovers the spatial size of the image from the deep representation while producing the segmented image as output. First, we fuse the information contained in the different HR modalities (optical, NIR, and elevation information for the FLAIR-2 dataset). For this purpose, we select a 2-D convolutional layer as the fusion layer. This fusion layer adapts the input size for the encoder, generating multimodal features with a low computational cost. Once these shallow multimodal features are generated, they are fed to the encoder blocks.

The mix vision transformer (MiT) [54] was chosen as the architecture for the encoder blocks. The MiT has a hierarchical encoder structure where deeper maps of different multilevel resolutions are extracted. Mixing HR and coarser-resolution feature maps allows for a better understanding of semantic classes. Transformer-based encoders are able to find mid- and long-range dependencies between features, which is fundamental for very HR images. The MiT architecture also uses overlapping patches and efficient self-attention to improve segmentation performance.

Then, the deep features are forwarded to a lighter decoder, where each block recovers the previous spatial size and reduces the features to generate the segmentation image. The decoder blocks are composed of 2-D convolution layers with batch normalization and rectified linear unit (ReLU) activation. Skip connections from each encoder output are also used in each decoder module as input to improve convergence and alleviate vanishing gradients. Finally, a segmentation head with another 2-D convolution is used to generate the class predictions.

2) Multi-Temporal Multispectral Network: For the multitemporal multispectral satellite images, we take the U-Net scheme with temporal attention proposed in [47], named U-TAE. This architecture uses sets of convolutional layers with batch normalization as encoder and decoder blocks to generate deep features and recover the original spatial size of each image of the time series. When each image is represented by the deepest features, the multi-temporal stack is fused with a multihead self-attention module. From the multi-temporal features, attention masks are generated for all the resolutions and included in the skip connections to the decoder. The attention masks add information about the temporal dependencies in the time series to each level of the network.

3) DTF Module: In order to fuse spatial and temporal features and interconnect both networks, we designed the DTF module. The DTF module interconnects both branches only at their deepest level, allowing precise spatial textures and changing temporal patterns to complement each other while preserving spatial detail and avoiding feature redundancy. The module is composed of two complementary blocks. A first block to align the spatial embeddings of both modalities

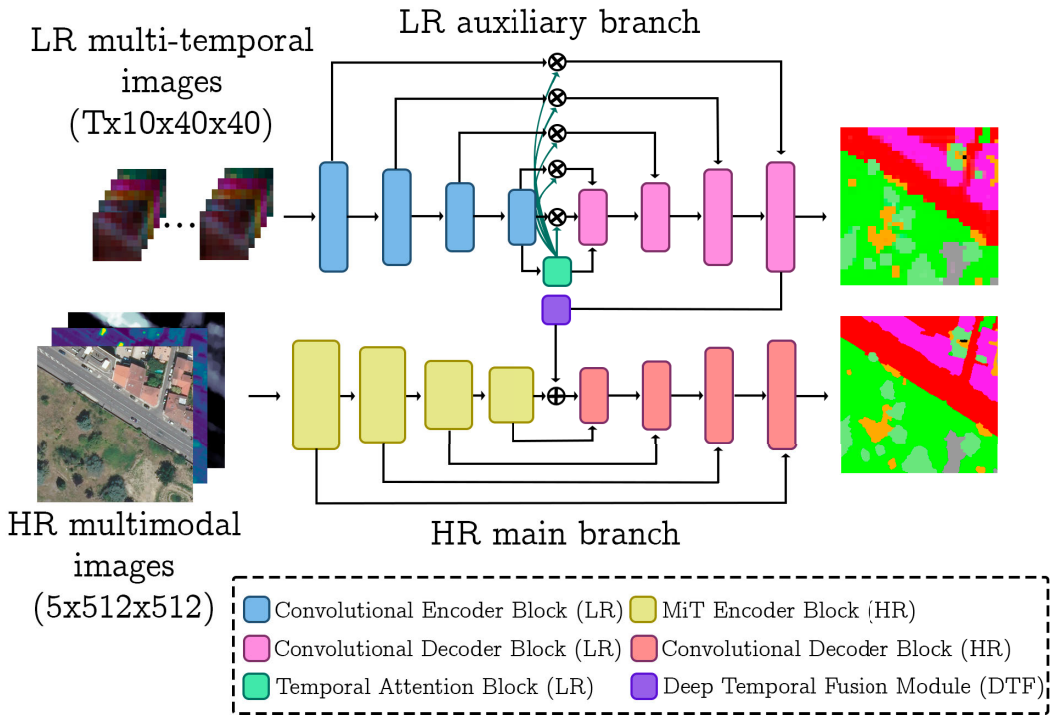


Fig. 2. Architecture of the DAM-Former. On the top, the LR multi-temporal branch, with the convolutional encoder and decoder blocks in blue and pink, respectively. Including the temporal attention block in green and the DTF module in purple. On the bottom, the HR multimodal branch, with the transformer encoders in yellow, and the lighter convolutional decoders in red.

and a second block that preserves surrounding contextual information. In the first block, a central crop CROP is performed to the deep temporal features F_t to match the information from the HR image. Then, the cropped features are processed through a convolutional layer L_C to match the dimensions of the HR feature map, and up-sampled through a bilinear resampling BR, ensuring the spatial extent of the HR and LR data and generating the cropped features F_s. The complete block process is defined as follows:

$$F_s = \mathrm{BR}(L_C(\mathrm{CROP}(F_t))). \tag{10}$$

In contrast, the second block starts by summarizing the spatial variations while retaining the temporal information through a spatial global averaging (11). Once the collapsed global features are obtained, they are fed to three dense layers, each with a dropout and ReLU activation (12). As the number of temporal samples might vary, the number of neurons N of these layers has to be dynamically adapted. For this purpose, a parameter m to adapt the neuron number in each layer is calculated from the input feature number C_{in} and the HR branch feature number C_{out}, as shown in (13). The value m is then added or subtracted from C_{in} depending on the case, following (14). In case of having the same number of features, an intermediate expansion with $C_{\text{in}} * 2$ neurons is done in the first two layers, and the third layer contains the final number of neurons C_{out}. After generating the spatial matching features F_s and the contextual global features F_g, they are added to the deepest features of the HR main branch F_d, as shown in (15), generating the new deep fused features from both branches F_{fd}

$$F_{g0} = \frac{1}{H \cdot W} \sum_{h=1}^{H} \sum_{w=1}^{W} F_t(h, w) \tag{11}$$

$$F_g = L_{d3}(L_{d2}(L_{d1}(F_{g0}))) \tag{12}$$

$$m = \frac{|C_{\text{in}} - C_{\text{out}}|}{3} \tag{13}$$

$$\begin{cases} N = [C_{\text{in}} + m, C_{\text{in}} + 2m, C_{\text{out}}], & \text{if } C_{\text{in}} < C_{\text{out}} \\ N = [C_{\text{in}} - m, C_{\text{in}} - 2m, C_{\text{out}}], & \text{if } C_{\text{in}} > C_{\text{out}} \\ N = [2C_{\text{in}}, 2C_{\text{in}}, C_{\text{out}}], & \text{if } C_{\text{in}} = C_{\text{out}} \end{cases} \tag{14}$$

$$F_{fd} = F_d + F_g + F_c. \tag{15}$$

IV. Dataset

In this work, we make use of the publicly available FLAIR-2 dataset, which provides great opportunities to experiment with domain adaptation techniques. The FLAIR-2 dataset contains multimodal images from 50 different French departments, covering 810 km^2 of the French metropolitan area. The dataset

$$\mathcal{L}_T(x_{T_A}, x_{T_B}, p_{T_e}) = -\sum_{i=1}^{H} \sum_{j=1}^{W} \sum_{c=1}^{C} \left(q_T^{(i,j)} \cdot p_{T_e}^{(i,j)} \right)_c \cdot \log \left(g_\theta \left(x_{T_A}^{(i,j)}, x_{T_B}^{(i,j)} \right) \right)_c \tag{9}$$

TABLE I
CLASSES AND ASSOCIATED COLORS FOR THE FLAIR-2 DATASET

Color	Class
	1. Building
	2. Pervious Surface
	3. Impervious Surface
	4. Bare Soil
	5. Water
	6. Coniferous
	7. Deciduous
	8. Brushwood
	9. Vineyard
	10. Herbaceous Vegetation
	11. Agricultural Land
	12. Plowed Land
	13. Others

contains 77 412 patches of several landscapes and climates, including different agricultural, forest, coastal, mountainous, and urban areas. The dataset is focused on aerial and satellite imagery for semantic segmentation, with 13 semantic classes of different soils, structures, and vegetation. The complete range of classes is shown in Table I.

The images composing the FLAIR-2 dataset have two main sources: aerial HR sensors and multi-temporal Sentinel-2 multispectral images. The aerial images were obtained between April and November in three different years (2018–2021). Differences in acquisition time for the HR images affect the appearance of some of the vegetation land cover classes, showing different stages of crop growth and seasonal changes in deciduous trees or forested areas. These images have a resolution of 0.2 m, with information in five bands, three optical RGB channels, an NIR band, and an elevation band generated by the difference between a DSM and a digital terrain model (DTM) from different sources. On the other hand, Sentinel-2 images have a spatial resolution of 10 m (20 m bands are resampled) and a spectral resolution of ten different bands, including visible and medium IR. Moreover, multi-temporal samples of each image are provided, including all the images acquired during the same acquisition year of the HR aerial image. However, the number of acquisitions varies for each image. This variety of multimodal information allows to differentiate between very similar classes in a multi-class segmentation task. For instance, elevation information distinguishes clearly between buildings and reflective soils and surfaces of similar colors; the NIR band provides information to differentiate between plant species and materials; and through multi-temporal samples, the difference between coniferous and deciduous plants can be observed as their foliage changes.

Even though the original FLAIR-2 dataset divides each department into a different domain, these domains are small enough to have neighboring domains with very similar climatic and terrain conditions. For this reason, in this work, we group and divide some of these domains into a North and a South domain. Specifically, we select the departments 80 (Somme), 60 (Oise), 08 (Ardennes), 51 (Marne), and 77 (Seine et Marne) for the North domain, and the departments 81 (Tarn), 32 (Gers), 31 (Haute Garonne), 46 (Lot), and 34 (Hérault) for the South domain. In Fig. 3, the areas covered by each domain are shown. The climate in the northern part of France is mostly affected by the Atlantic Ocean, with very high humidity and rains, cold winters, and moderately warm summers. This climate, combined with the wide plain terrain dominating the north, has grasslands and highly productive and fertile lands for crops such as wheat, barley, or corn. Conversely, in the southern regions of the country, the Mediterranean climate is characterized by hot and dry summers and mild, wet winters. At the same time, the southern and eastern parts of the country have much more rugged terrain, with more forests of oaks, pines, and brushwood. In this part of the country, vineyards, olive trees, orchards, and horticulture are predominant. In Fig. 4, the specific distribution of classes for the North and South domains is shown. It can be seen that, on the one hand, the North domain contains a higher number of urban classes, such as buildings, pervious and impervious surfaces, grasslands of herbaceous vegetation and plowed lands, and coastal and maritime areas with water bodies. On the other hand, the South domain contains far more vineyard areas, deciduous trees such as different species of oaks, and bare soil.

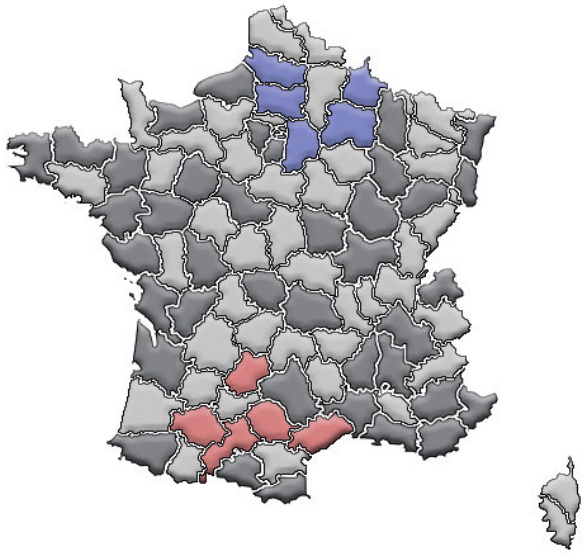


Fig. 3. Map of the metropolitan region of France, with the departments covered in the North domain in blue, in the South domain in red, and dark gray for other departments included in the FLAIR-2 dataset.

V. EXPERIMENTS

In this section, we include different experimental results to evaluate the performance and effectiveness of the proposed method for performing UDA from the North to the South domain and from the South to the North domain. First, the architecture details and hyperparameters, as well as the training settings, are defined in Section V-A. Then, an extensive class analysis with optical, multimodal, and temporal information for different architectures is performed in Section V-B. We analyze the results of domain adaptation tasks from the North to the South domain and vice versa, including quantitative and qualitative assessments. Then, a detailed ablation study of the model and its modules (see Section V-C) as well

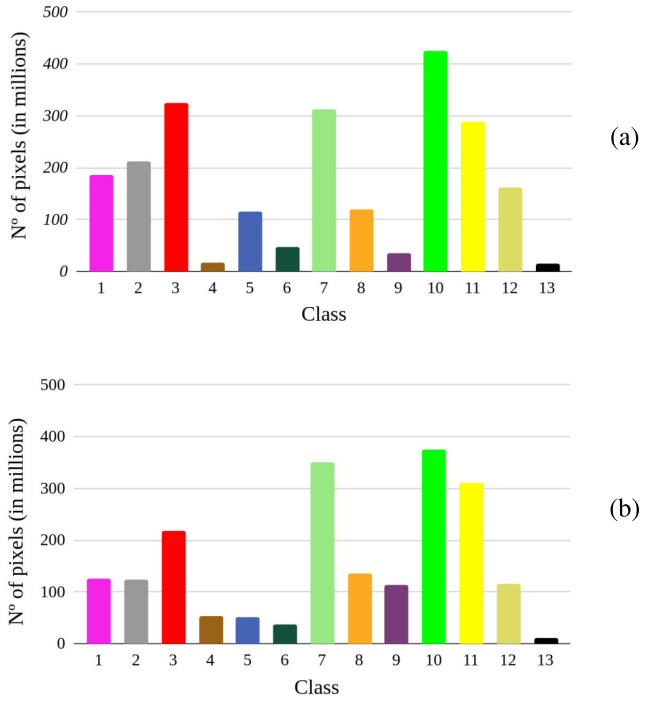


Fig. 4. Class distribution for each domain generated from the FLAIR-2 dataset: (a) North domain and (b) South domain.

as the upper-bound (Oracle) analysis of the proposed model and other architectures (see Section V-D) are examined.

A. Experimental Settings

The proposed multimodal UDA model was implemented in Python3.8 using Pytorch on an Nvidia Tesla V100 and a CPU Intel[1] Xeon[1] E5-2698 v4. For the architecture parameters, in the HR network, we used a fusion convolution layer with 5 input channels, including the RGB, NIR, and elevation bands from aerial FLAIR-2 images, and an output of three channels with a kernel size of 1 to adapt the input data to the MiT encoders. For the encoders, we select the MiT-B4 pretrained on the ImageNet dataset [82] as the backbone, with embedding dimensions of [64, 128, 320, 512], with [1, 2, 4, 8] heads and [3, 8, 27, 3] depths. The decoder blocks' convolutional layers have the same dimensions as the corresponding encoder blocks, [64, 128, 320, 512] with a kernel size of 3×3. For our temporal network, we use an U-TAE network of 4 levels, with [64, 64, 128, 128] feature map sizes for each level, and 16 heads and 256 dimensions for the temporal attention module.

During the training of the networks, every model was trained with a batch size of 2 during 40 000 iterations. We follow the standard hyperparameter selection for our optimizer, RCS and DACS of previous works. With an AdamW optimizer with a learning rate of $6 \cdot 10^{-5}$, 0.9 and 0.999 betas, and 0.01 weight decay. A linear learning rate warm-up ratio r_{warm} of 10^{-6} for the first 1500 warm-up iterations T_{warm} with subsequent polynomial decay was also adopted. The RCS was applied to every sample with less than 3000 pixels of a class (RCS_{th}, a class temperature RCS_T of 0.01, and a minimum crop ratio of 0.5. The color jitter strength and probability for the augmentation were set to 0.2. The hyperparameters for the pseudo-label ensemble were set to $\beta = 1.0$ and $\gamma = 0.2$, increasing the weight of the predictions from the HR branch. The threshold value for the quality function q_T was set as $\tau = 0.968$ and the teacher momentum α to 0.99. In Table II the hyperparameters are summarized.

[1]Registered trademark.

TABLE II
EXPERIMENTAL SETTINGS

Hyperparameter	Value
Iterations (N)	40000
Batch size	2
Learning rate (η)	$6 \cdot 10^{-5}$
AdamW Betas	[0.9, 0.999]
Weight decay	0.01
Warm-up iterations (T_{warm})	1500
Warm-up ratio (r_{warm})	10^{-6}
RCS threshold (RCS_{th})	3000
RCS temperature (RCS_T)	0.01
RCS crop	0.5
Color jitter strength	0.2
Color jitter probability	0.2
Pseudo-label β	1.0
Pseudo-label γ	0.2
Quality function (τ)	0.968
Teacher momentum (α)	0.99
Seed	0

B. General UDA Segmentation Results

Extensive experimentation is required in order to better analyze our model and the problem of UDA in cases where complex multimodal data can be exploited, such as the FLAIR-2 dataset. For this purpose, both the NS and SN domain adaptation cases are considered. To analyze the segmentation results for each class in different environments, we first use models employing only optical (RGB) information. Then, we add NIR and elevation information to the models (NIR + E). After this, we mix both of these cases with multi-temporal information. We also evaluate different models only using multi-temporal information. To evaluate these experiments, for each class, we employ the IoU to quantify the predicted segmentation region and the accuracy (Acc) for individual pixel prediction. For overall segmentation results, we consider the all-pixel accuracy (AAcc), which measures the percentage of accurately predicted pixels in the complete image; the mean pixel accuracy (MAcc), which is the mean of correctly segmented pixels for each class; and the mIoU, which indicates the mean precisely segmented areas over the image. Then, we interpret several visual examples for each method compared, as well as analyze the confusion matrix for the multimodal multi-temporal models.

1) UDA Segmentation, Only HR Information: First, we perform the UDA with three different popular network architectures, using as input only the RGB information from the HR images. Specifically, we first include a convolutional U-Net [U-Net (C)] [5] network, which was proposed as the baseline in previous image segmentation works for the FLAIR-1 [83]

TABLE III

QUANTITATIVE PERFORMANCE OF MODELS WITHOUT TEMPORAL INFORMATION IN TERMS OF CLASS ACCURACY (ACC) AND INTERSECTION OVER UNION (IOU), AS WELL AS ALL PIXEL ACCURACY (AACC), MEAN ACCURACY (MACC), AND MEAN IOU (MIOU) ON THE FLAIR-2 DATASET FROM NORTH TO SOUTH (NS). THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND-BEST RESULT UNDERLINED

Class	RGB						RGB+NIR+E					
	U-Net (C)		DAFormer		U-Net (T)		U-Net (C)		DAFormer		U-Net (T)	
N → S	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	41.04	44.98	31.69	36.43	53.20	70.05	75.68	88.37	74.18	88.01	**77.34**	**92.13**
P.Surface	18.72	31.40	18.04	30.69	**70.05**	**49.11**	14.51	18.39	24.34	40.09	22.89	28.77
I.Surface	45.33	61.91	44.96	63.09	31.81	65.84	57.87	68.87	55.49	63.37	**62.80**	**73.65**
B.Soil	5.52	7.44	14.06	**37.89**	**48.62**	10.45	7.80	9.42	4.53	15.94	4.52	8.68
Water	21.62	25.02	26.54	68.08	8.59	77.47	63.01	79.55	63.27	86.20	**63.97**	**90.51**
Coniferous	0.08	0.08	12.39	27.13	**57.04**	45.87	23.84	47.25	21.54	40.78	24.42	**52.60**
Deciduous	50.54	68.88	52.57	70.15	26.48	71.02	64.45	78.11	59.69	76.45	**65.37**	**78.99**
Brushwood	1.52	1.63	9.97	14.76	**53.75**	20.87	18.90	24.81	16.42	27.73	22.85	**32.77**
Vineyard	2.37	3.03	9.33	1.05	15.45	5.15	41.91	48.26	13.34	21.79	**64.28**	**73.28**
H.Vegetation	31.52	**73.97**	30.60	56.47	4.72	58.65	33.50	54.99	32.29	50.46	**40.99**	68.40
A.Land	31.42	48.07	18.69	23.86	32.68	56.66	38.85	**71.91**	31.76	44.54	**44.01**	64.79
P.Land	11.15	12.22	31.97	48.51	**41.10**	**59.11**	23.49	30.32	6.77	7.62	2.53	2.75
Others	0.62	2.48	0.83	3.48	0.87	**4.74**	0.87	2.17	0.84	4.40	**1.31**	2.53
AAcc	46.19		44.83		54.50		58.05		51.10		**61.71**	
MAcc	29.32		37.82		45.77		47.88		43.64		**51.53**	
mIoU	20.11		23.20		32.28		35.75		31.11		**38.25**	

TABLE IV

QUANTITATIVE PERFORMANCE OF MODELS WITHOUT TEMPORAL INFORMATION IN TERMS OF CLASS ACCURACY (ACC) AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET FROM SOUTH TO NORTH (SN). THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND-BEST RESULT UNDERLINED

Class	RGB						RGB+NIR+E					
	U-Net (C)		DAFormer		U-Net (T)		U-Net (C)		DAFormer		U-Net (T)	
S → N	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	26.62	29.03	15.02	16.08	44.07	56.79	61.18	72.39	65.48	77.05	**71.79**	**82.73**
P.Surface	7.50	8.90	20.98	26.04	**34.17**	**45.08**	18.17	28.36	27.84	38.08	30.72	38.89
I.Surface	28.00	33.91	43.29	67.86	44.71	61.56	50.71	69.91	49.12	59.00	**59.32**	**70.67**
B.Soil	5.98	51.58	6.35	46.18	**13.60**	48.97	6.91	51.92	6.08	42.64	12.02	**60.55**
Water	23.24	74.48	36.95	60.06	55.09	74.37	67.94	88.33	65.00	87.91	**81.76**	**92.44**
Coniferous	14.38	15.88	2.28	2.57	28.60	36.86	27.98	48.60	17.36	**57.10**	**33.76**	48.37
Deciduous	35.58	44.04	43.22	60.38	47.87	69.76	60.30	71.04	44.43	51.02	**65.77**	**81.40**
Brushwood	0.12	0.12	15.11	35.30	23.50	50.35	23.96	38.86	19.48	**60.68**	**27.42**	52.26
Vineyard	0.16	0.17	**29.88**	**52.08**	1.30	1.39	0.09	0.11	5.71	17.07	7.33	9.48
H.Vegetation	28.20	**70.08**	30.51	46.86	30.14	43.48	36.72	60.81	31.20	43.50	**40.25**	55.65
A.Land	15.14	18.68	17.33	24.36	**35.25**	53.07	23.48	31.73	20.09	24.97	33.79	**54.86**
P.Land	10.71	11.43	40.94	54.95	**46.30**	**64.56**	26.47	32.88	30.27	42.72	32.49	45.50
Others	0.76	**8.47**	0.56	4.87	1.02	1.19	0.21	0.42	0.26	0.88	**1.93**	3.36
AAcc	35.17		43.90		54.60		54.32		49.38		**61.72**	
MAcc	28.21		38.28		46.72		45.80		46.36		**53.55**	
mIoU	15.11		23.26		31.20		31.09		29.41		**38.34**	

and FLAIR-2 [42] datasets. This network has a ResNet34 [84] backbone, pretrained on the ImageNet dataset [82]. Another popular architecture from recent times is the DAFormer [37], a transformer-based architecture with encoders for different resolutions without positional encoding and a very light decoder. The MiT-B4 was selected as the backbone [54] for the DAFormer. Similar approaches can be seen in remote sensing applications for UDA [36]. Then, we introduce the architecture proposed for our HR multimodal network, the DAM-Former, a U-Net scheme with a transformer-based encoder and a convolutional decoder, referred to as U-Net (T), further details are given in Section III-C.

In Tables III and IV, the NS and SN domain adaptation segmentation results are shown. The same architectures using the NIR + E information are also included in the same table to evaluate and compare the effects of the multimodal information across the different classes. In both tables, it can be seen how the methods that include the NIR + E information significantly increase their performance compared to their RGB-only counterparts. In general, the best segmentation results for both UDAs are obtained by the U-Net (T), especially the model using multimodal NIR + E information.

For the NS UDA in Table III, there are some trends that can be seen across the different models. First, the segmentation results for the classes "bare soil," "plowed land," and "others" are generally low. The classes that most benefit from the NIR + E information are the vegetation classes such as "vineyard" and "brushwood," as their height and NIR information can be representative. From the models using only the RGB images, the U-Net (C) results were the lowest, with several classes underrepresented. Even classes where other methods manage to classify to some extent, for example, the

TABLE V

QUANTITATIVE PERFORMANCE OF MODELS WITH ONLY THE SENTINEL-2 MULTI-TEMPORAL INFORMATION IN TERMS OF CLASS ACCURACY (ACC) AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND-BEST RESULT UNDERLINED

Class	N → S						S → N					
	3DCNN		LSTM		U-TAE		3DCNN		LSTM		U-TAE	
	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	0.02	0.02	11.47	13.76	**31.05**	**50.28**	0.00	0.00	0.04	0.04	**17.34**	**21.66**
P.Surface	5.74	7.95	9.39	13.13	**22.09**	**32.66**	0.00	0.00	11.84	14.38	**26.28**	**39.26**
I.Surface	**29.61**	**61.61**	27.91	56.07	28.17	44.85	21.00	31.18	27.02	49.60	**33.73**	**56.37**
B.Soil	1.76	1.90	**26.78**	**32.89**	9.88	10.97	6.76	27.92	8.78	**43.48**	**9.62**	30.03
Water	25.86	30.85	33.59	45.29	**55.07**	**74.71**	17.12	18.94	35.14	57.90	**61.04**	**78.04**
Coniferous	11.30	13.48	**21.74**	**30.32**	18.17	24.54	11.30	17.05	20.32	28.25	**29.78**	**36.76**
Deciduous	16.88	19.56	19.01	24.26	**37.21**	**45.25**	22.95	**77.57**	25.96	62.86	**37.43**	74.07
Brushwood	3.25	3.79	1.42	1.63	**5.62**	**8.22**	0.11	0.12	5.54	7.34	**13.46**	**33.31**
Vineyard	3.89	4.46	0.64	0.91	**6.82**	**10.79**	0.00	0.00	0.03	0.03	**8.44**	**12.60**
H.Vegetation	21.62	**72.39**	19.89	54.91	**28.62**	69.96	17.56	**36.81**	18.30	33.77	**20.16**	26.87
A.Land	9.47	11.34	14.09	18.88	**25.14**	**33.67**	18.83	29.39	23.58	35.74	**32.59**	**50.50**
P.Land	0.00	0.00	**19.58**	**36.23**	11.36	14.67	0.00	0.00	**11.88**	**13.59**	5.63	7.00
Others	**1.12**	**12.48**	0.01	0.03	0.11	0.23	**0.62**	**1.22**	0.10	0.16	0.46	0.63
AAcc	27.30		29.95		**40.56**		27.42		33.33		**42.74**	
MAcc	18.45		26.02		**32.37**		18.48		26.70		**35.93**	
mIoU	10.04		15.81		**21.48**		8.94		14.50		**22.77**	

TABLE VI

QUANTITATIVE PERFORMANCE OF MODELS WITH TEMPORAL INFORMATION IN TERMS OF CLASS ACCURACY (ACC) AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET FROM NS. THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND-BEST RESULT UNDERLINED

Class	RGB+TEMPORAL						RGB+NIR+E+TEMPORAL					
	U-Net (C)		DAFormer		U-Net (T)		U-Net (C)		DAFormer		DAM-Former	
N → S	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	53.67	72.29	20.66	26.96	51.45	71.07	76.33	85.57	72.36	90.35	**80.05**	**90.40**
P.Surface	20.59	**54.66**	27.58	49.66	16.66	18.61	27.65	47.29	26.91	53.01	**28.77**	37.36
I.Surface	44.52	57.75	30.37	45.23	44.77	61.45	55.50	64.06	57.82	69.67	**65.74**	**74.79**
B.Soil	3.86	4.22	5.43	**19.24**	3.30	3.64	5.75	6.87	**8.29**	9.99	4.29	7.37
Water	63.53	73.42	58.15	81.18	45.60	51.66	**71.55**	90.03	71.38	86.38	63.39	**91.89**
Coniferous	20.64	26.63	6.03	6.21	13.98	18.39	**31.93**	**61.92**	28.38	48.40	31.58	53.00
Deciduous	49.32	58.20	37.20	75.76	50.12	61.71	62.82	72.38	64.07	80.75	**68.21**	**82.60**
Brushwood	6.37	8.85	14.04	**23.80**	2.50	2.58	13.08	15.07	13.80	19.41	**24.63**	21.42
Vineyard	2.43	2.89	0.55	0.56	1.10	1.49	12.77	16.76	8.80	10.41	**61.30**	**71.54**
H.Vegetation	29.72	71.67	23.52	38.45	28.50	**79.94**	33.71	57.63	40.34	70.08	**43.33**	70.97
A.Land	21.01	25.69	26.30	39.88	26.57	30.77	37.66	**73.36**	33.32	40.95	**47.26**	67.63
P.Land	3.07	3.46	0.75	0.76	31.26	35.05	6.49	7.21	**47.41**	**68.30**	18.54	22.63
Others	0.21	0.70	0.26	0.37	0.76	**3.83**	0.57	1.47	0.71	2.02	**1.37**	2.91
AAcc	44.79		41.91		46.42		55.59		58.78		**64.75**	
MAcc	35.42		31.39		33.86		46.12		49.98		**54.19**	
mIoU	24.53		19.29		24.35		33.52		36.43		**41.42**	

"coniferous" class. The DAFormer obtained more balanced results; however, most of its results are not competitive. For the multimodal models, it can be observed that the segmentation results from the U-Net (C) and the DAFormer are much closer, with the U-Net (C) on top. Nevertheless, the results obtained by the U-Net (T) are better for most classes.

For the SN UDA results from Table IV, the "bare soil" and "others" classes were also the least accurately classified. Compared to the previous DA, the "vineyard" class suffers more from misclassification in most of the models. In this case, most of the models have lower performance than in the North-South case. Specifically, the U-Net (C) for both RGB and NIR + E information. The U-Net (T), however, was not affected as much, but the accuracy class distributions were also different, with better results for "bare soil" or "plowed land" and worse results for "vineyard." The performance drop in "bare soil" and "vineyard" was expected due to their lower representation in the dataset. For other classes, such as "plowed land," further analysis is performed in Section V-B5.

2) UDA Segmentation Results, Only Temporal Information: Second, we analyze the performance of three different networks over the multi-temporal and multispectral images. For this purpose, we compare the chosen model for the temporal branch, the U-TAE [47], with two other popular models. The first compared model is a shallow U-Net with 3-D convolutional layers [85] (3D-CNN) of four levels. The second model combines a bidirectional convolutional long short-term memory (LSTM) with a feature pyramidal network [86]. The results included in Table V show that the U-TAE is able to obtain the best results, followed by the LSTM, and finally the 3DCNN. Especially in the SN DA, the U-TAE was able to outperform the other models in most of the classes, except for the Plowed Land and Others classes. After this experiment,

TABLE VII

QUANTITATIVE PERFORMANCE OF MODELS WITH TEMPORAL INFORMATION IN TERMS OF CLASS ACCURACY (ACC) AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET FROM SN. THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND BEST RESULT UNDERLINED

Class	RGB+TEMPORAL						RGB+NIR+E+TEMPORAL					
	U-Net (C)		DAFormer		U-Net (T)		U-Net (C)		DAFormer		DAM-Former	
S → N	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	19.62	22.00	30.39	51.74	38.46	43.95	56.01	59.40	61.37	69.37	**72.53**	**86.27**
P.Surface	22.69	29.01	17.53	29.37	21.10	23.46	26.97	**46.54**	**30.98**	40.88	30.39	38.12
I.Surface	34.85	47.70	31.49	52.74	31.52	34.12	48.43	66.64	52.25	71.26	**61.35**	**75.02**
B.Soil	6.66	22.67	8.65	11.01	**16.19**	54.11	10.63	45.82	11.28	47.12	11.24	**54.22**
Water	59.31	79.77	60.14	73.10	26.17	26.63	70.94	91.82	67.38	**93.51**	**84.98**	91.93
Coniferous	25.27	29.62	27.89	**50.55**	18.56	20.72	**36.25**	45.53	28.51	31.64	33.69	49.27
Deciduous	41.44	58.07	38.84	49.35	33.78	39.29	64.74	79.80	61.33	**81.78**	**64.86**	79.90
Brushwood	13.68	17.72	2.94	3.79	21.25	28.68	25.86	44.45	22.12	**58.30**	**27.07**	48.59
Vineyard	0.34	0.46	2.39	2.71	6.85	7.11	1.55	1.73	1.20	1.70	**40.62**	**49.76**
H.Vegetation	28.83	78.02	27.84	67.78	24.95	**88.37**	36.70	57.31	35.00	55.46	**42.96**	59.92
A.Land	19.36	23.69	22.43	28.68	13.54	14.37	30.81	48.29	26.73	22.29	**37.93**	**61.07**
P.Land	0.58	0.60	13.61	17.28	**32.13**	**35.01**	14.66	16.96	24.10	27.52	23.52	30.66
Others	0.50	1.45	0.41	1.04	0.11	0.12	**0.88**	**1.08**	0.21	0.30	0.54	0.63
AAcc	42.98		40.69		40.95		56.39		56.66		**63.23**	
MAcc	31.60		33.78		32.00		46.57		47.09		**55.80**	
mIoU	21.01		21.89		21.89		32.65		32.50		**40.90**	

confirming the performance of the U-TAE for the Sentinel-2 multi-temporal data, we proceeded to use the U-TAE network as the temporal branch for the following experiments.

3) UDA Segmentation Results, Multimodal Information: In third place, in order to benefit from the complete information supplied by the FLAIR-2 dataset, the multi-temporal information from Sentinel-2 images are included. For this reason, and in order to better evaluate and compare the proposed method, we include the same previously seen variations in Section V-B1 with temporal information. A temporal branch based on the best-performing model from Section V-B1, the U-TAE architecture, was used for all the methods. For the U-Net (C) and U-Net (T), we included the same temporal fusion scheme as proposed in [42]. For the DAFormer, the temporal features are concatenated with the fine-to-coarse features from the encoders in concordance with its general scheme. In Tables VI and VII, the results, including temporal information, for these methods are shown. First, it can be seen that the transformer-based models using RGB and temporal information are not improving their performance compared to their RGB counterparts. Moreover, some of these methods, including the U-Net (T), had a performance drop in comparison. When the NIR + E information is included, the performance is not increased in all the models, particularly the U-Net (C). This suggests that just including the temporal information in the system does not translate into a performance increase for UDA segmentation tasks, even for an architecture specially designed for the FLAIR-2 dataset image segmentation task.

In the NS case, shown in Table VI, in most of the cases, the proposed method outperforms all the other methods. Except for the "plowed land" class, where it obtains a close result to the temporal DAFormer, our model is able to obtain better results, especially in the "vineyard" class. Regarding the other methods, methods without NIR + E information do not stand out in any category with a relevant margin compared to their RGB-only counterparts. For the methods with NIR + E modalities, the transformer-based temporal DAFormer is able to obtain better accuracy performance for different soil classes, such as "pervious surface," "impervious surface," "bare soil," and "plowed land," while the U-Net (C) has better results

TABLE VIII

QUANTITATIVE PERFORMANCE ABLATION STUDY OF THE PROPOSED MODEL AND ITS DIFFERENT MODULES: MIXED TEMPORAL LOSS (ML), DTF, AND THE EPS IN TERMS OF ALL PIXEL ACCURACY (AACC), MACC, AND MIOU ON THE FLAIR-2 DATASET. THE BEST RESULTS ARE SHOWN IN BOLD, WITH THE SECOND-BEST RESULT UNDERLINED

Model	N → S			S → N		
	AAcc	MAcc	mIoU	AAcc	MAcc	mIoU
ML	62.46	52.40	38.03	61.73	54.68	39.34
ML + DTF	63.44	52.63	40.54	**63.67**	**56.84**	**40.90**
ML + EP	61.50	50.82	37.54	60.68	52.72	38.00
ML + DTF + EP	**64.75**	**54.19**	**41.42**	63.23	55.80	**40.90**

TABLE IX

QUANTITATIVE PERFORMANCE RESULTS OF DIFFERENT TEMPORAL FUSION APPROACHES, IN AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET. THE BEST RESULTS ARE SHOWN IN BOLD

Fusion Approach	N → S			S → N		
	AAcc	MAcc	mIoU	AAcc	MAcc	mIoU
All Features	56.96	47.12	32.77	58.65	50.50	35.52
Deep Features	**63.44**	**52.63**	**40.54**	**63.67**	**56.84**	**40.90**

TABLE X

MODEL COMPLEXITY COMPARISON

Model	Parameters
3D-CNN	1.6M
LSTM	1.7M
U-TAE	2.3M
U-Net (C)	24.4M
U-Net (T)	64.1M
DAFormer	85.2M
U-Net + Temporal (C)	27.3M
U-Net + Temporal (T)	66.4M
DAFormer + Temporal	89M
DAM-Former	67M

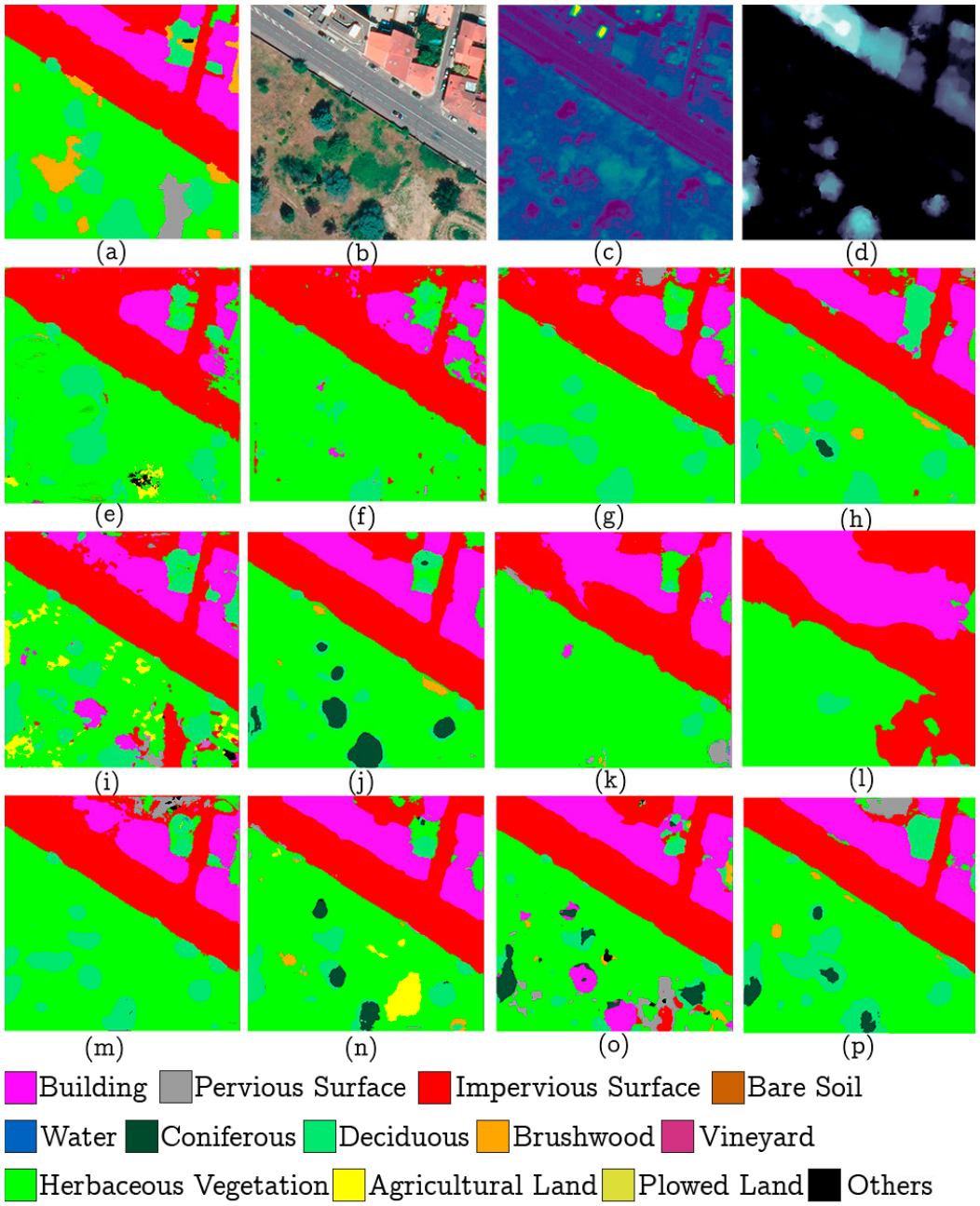


Fig. 5. NS UDA segmentation visual results covering a suburban area with a large portion of different vegetation, roads, and buildings, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

for vegetation classes such as "coniferous" or "vineyard." Nevertheless, the proposed DAM-Former is able to obtain around 5% better results in all the overall metrics. Compared to the next better-performing model, the multimodal U-Net (T) with NIR-E information, our model obtained more than a 3% improvement in all the general metrics. For the rest of the models, the differences range between 5% and 20%.

Similarly, for the SN UDA segmentation results presented in Table VII, the proposed method also outperforms the rest of the architectures. In this case, however, the Acc and IoU metrics for the individual classes are more balanced. Only the "others" class has a low prediction performance, and the "bare soil" class is the only IoU classification result with less than 20% of IoU. In general, compared to the models without temporal information, the results of classes such as "vineyard" or "bare soil" seem to be better classified thanks to the multi-temporal Sentinel-2 images. Nevertheless, as it could be seen in the previous NS UDA, the RGB transformer-based models with temporal information perform worse than their only RGB mode versions. For the models with NIR and elevation information, a similar trend as in the previous case can be seen, yet the increase in performance is lower than for the previous NS UDA task. For the SN UDA, the proposed method obtained more than a 2% performance increase compared to

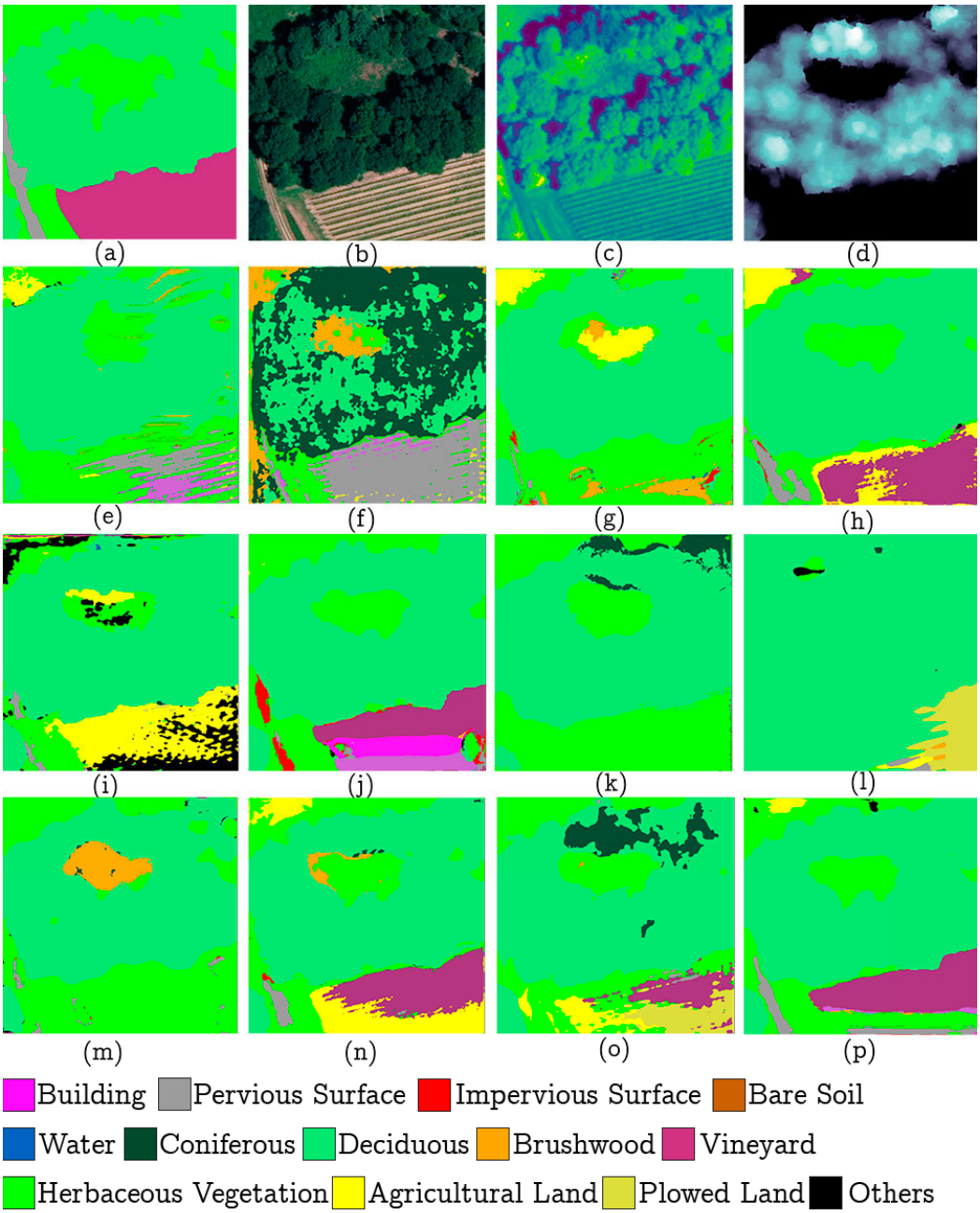


Fig. 6. NS UDA segmentation visual results covering an agricultural area with vineyards and different kinds of vegetation, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

the next best model (U-Net (T) with multimodal information), but an increase of more than a 60% in accuracy and 50% of IoU can be observed for the "vineyard" class. Compared to the rest of the models, the proposed model achieves a general improvement from 8% to 25% in the overall metrics.

4) UDA Segmentation Results, Qualitative Visual Examples: After the quantitative class analysis, three visual examples for each UDA segmentation task, NS and SN, have been included. In order to analyze and better understand previous results, each sample includes a different landscape with classes that obtained low accuracy in the previous experiments. In Fig. 5, a mix of urban and rural areas from the NS UDA is shown. In this case, the proposed model is able to perform an accurate segmentation of the buildings and other urban classes, except for the "previous surface" in the vegetation. The DAM-Former is able to correctly classify most of the vegetation, even though it confuses the highest parts of the deciduous trees with coniferous trees. In a similar way, it detects that there is "brushwood," but not to its complete extent. In comparison, most of the other methods contain high levels of noise in the vegetation, particularly the DAFormer-based models. In the remaining models, we can observe that the "coniferous" class

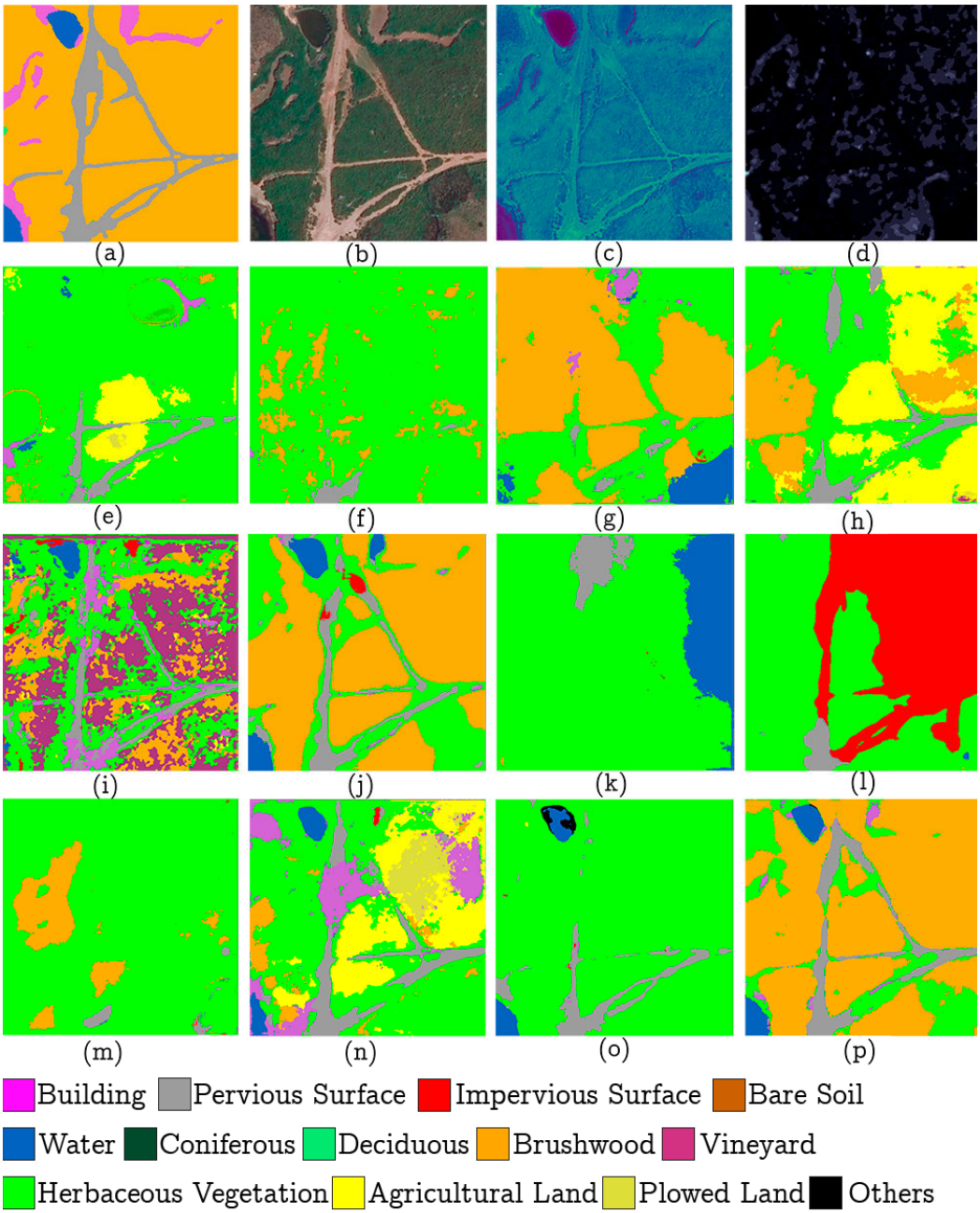


Fig. 7. NS UDA segmentation visual results covering a rural area with dirt roads, brushwood, crops, and water bodies, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

is even more misclassified, as in the U-Net (T) with NIR + E information, or that the urban classes are very deformed, as in the U-Net (C) with NIR + E information. In Fig. 6, an agricultural land with vineyards and some trees is shown. In this case, it can be seen how very few models are able to properly classify the "vineyard" class. At the same time, it can be seen that the proposed model is the only one able to classify most of the vineyard, deciduous trees, and roadway. Meanwhile, most of the models confuse the vegetation classes or completely ignore the roadway. As a last example for the NS UDA, in Fig. 7, a rural road surrounded by brushwood and a couple of water bodies is included. In this case, we can see how only a few models can classify the brushwood or the road shape correctly. However, only the proposed model and the U-Net (T) with NIR + E information are able to generate a close result to the ground truth, with the proposed method being able to better define the road edges.

For the South-to-North domain adaptation, three different landscapes were chosen. Fig. 8 shows a suburban area with a road, buildings, and some vegetation. In this case, it can be seen how most of the methods fail to classify the "coniferous," "herbaceous vegetation," "building," or "pervious surface"

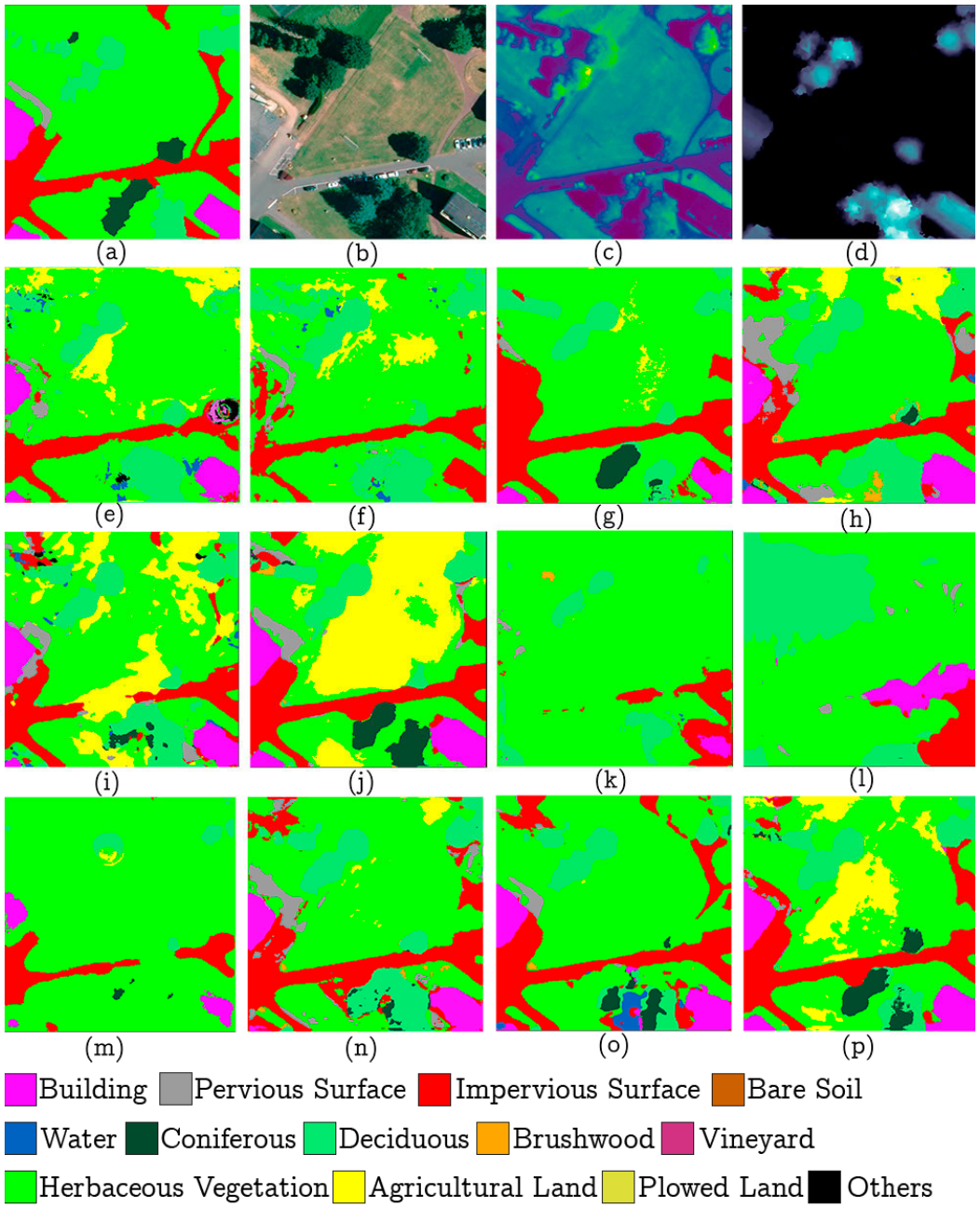


Fig. 8. SN UDA segmentation visual results covering a suburban area with grassland, coniferous and deciduous vegetation, roads, and buildings, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

classes. Even though the proposed model classifies some of the "herbaceous vegetation" as "agricultural land," it is able to differentiate between coniferous and deciduous trees much better than the other models while still classifying the urban infrastructure classes correctly with accurate borders. Among the other models, only the U-Net (T)-based models are able to find the coniferous trees. Nevertheless, their predictions are less sharp, and other classes are more confused. In Fig. 9, a different scenario is shown, with a highly urbanized area with a river and a bridge. In the predicted results, it can be observed that several models cannot properly classify the river or its shore. Several models also contain high amounts of noise. However, thanks to the proposed noise reduction techniques, such as the EPs, the proposed DAM-Former is able to generate a much smoother image with still sharp edges for many structures. It is interesting to note that many models classified the bridge as "building," probably because of its height, as can be seen in the elevation image. In the last included figure, Fig. 10, a suburban area can be seen, including many different classes. Most of the models are able to classify roads and buildings broadly. Nevertheless, the proposed model is the only one able to classify the vineyards in the image, most

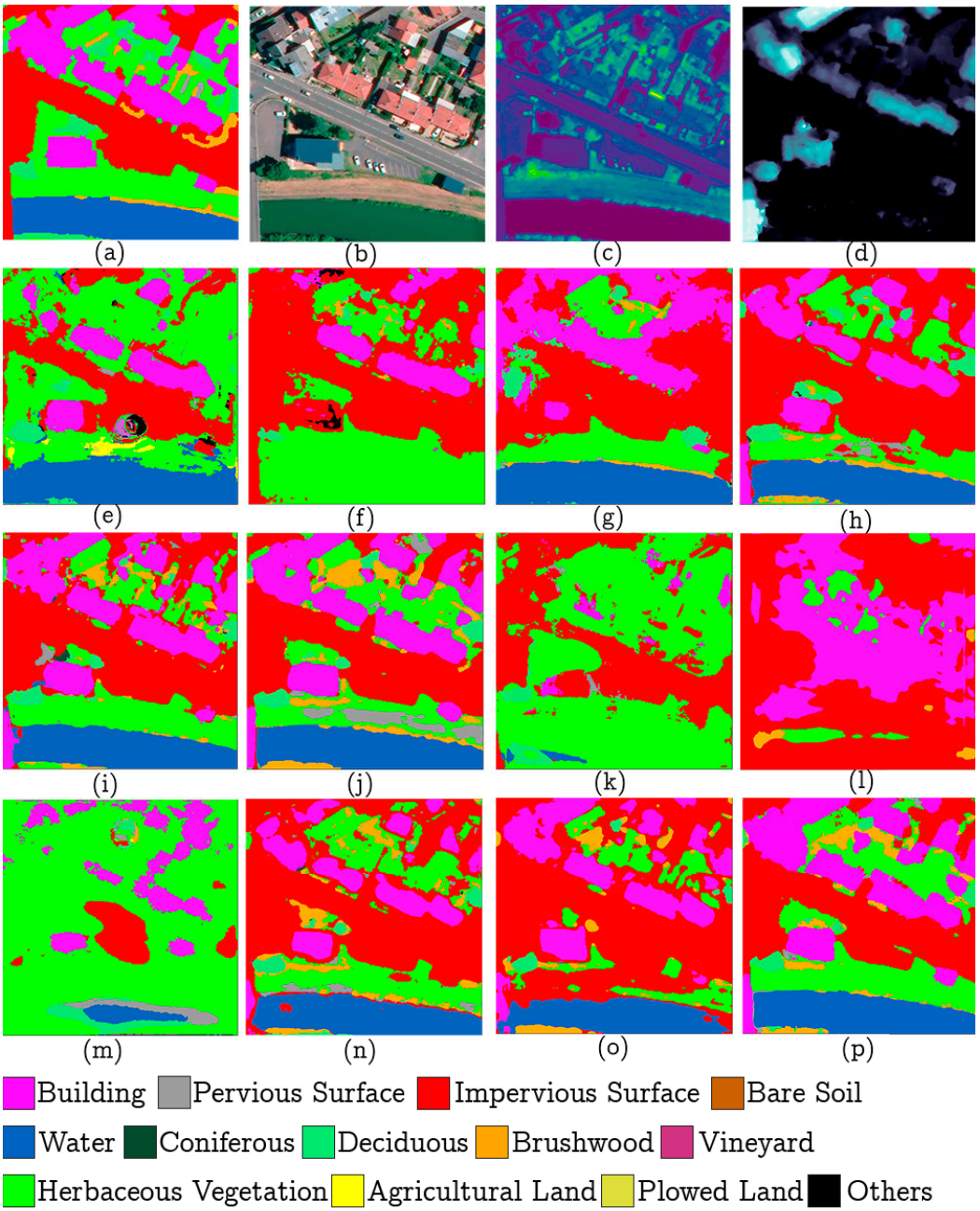


Fig. 9. SN UDA segmentation visual results covering an urban area with a large number of buildings, roads, parks, and a river, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

of the brushwood and other vegetation, and even the sparse previous surfaces.

5) UDA Segmentation Results, Confusion Matrices: In the previous experiment, some of the classes were wrongly classified and confused with others. To better understand this phenomenon, the confusion matrices for the models with temporal and HR multimodal information have been evaluated in Fig. 11. The first three matrices correspond to the North-to-South UDA, the U-Net (C), DAFormer, and the proposed model, respectively. In the first two, it can be seen that the "bare soil" (4) class has been mostly confused with "previous surface" (2), while the proposed model confused it with "herbaceous vegetation" (10). Similarly, while the U-Net (C) and DAFormer confuse the "others" (13) class with "pervious surface" (2) as well, for the proposed model, it is mostly confused with "bare soil" (4). It can also be seen how the proposed model performs much better, not confusing the "vineyard" (9) and "brushwood (8) classes. In the vegetation classes, the "coniferous" (6) class is mostly confused with the "deciduous" (7) class, as both of them are composed of high trees, which can be difficult to differentiate. In general, it can be seen that for the first two models, several classes

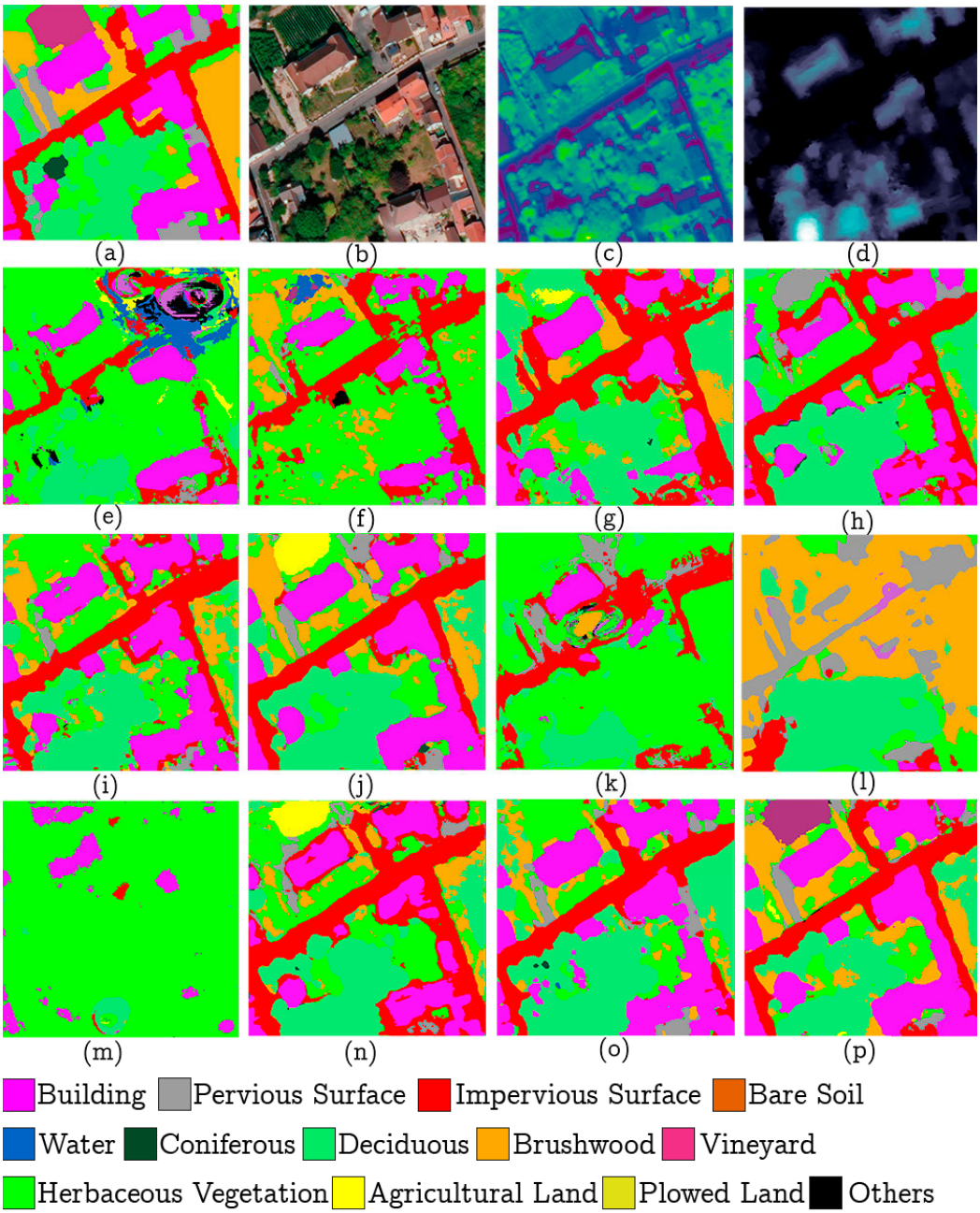


Fig. 10. SN UDA segmentation visual results covering a rural and agricultural area with vineyards, buildings, and different materials and vegetation, where: (a) ground truth, (b) RGB image, (c) NIR image, (d) elevation image, (e) RGB U-Net (C), (f) RGB DAFormer, (g) RGB U-Net (T), (h) RGB + NIR + E U-Net (C), (i) RGB + NIR + E DAFormer, (j) RGB + NIR + E U-Net (T), (k) RGB + T U-Net (C), (l) RGB + T DAFormer, (m) RGB + T U-Net (T), (n) RGB + NIR + E + T U-Net (C), (o) RGB + NIR + E + T DAFormer, and (p) RGB + NIR + E + T **DAM-Former**.

are wrongly labeled as "pervious surface" (2) or "herbaceous vegetation" (10). Meanwhile, in the proposed model, only the overfitting to "herbaceous vegetation" (10) can be seen in some classes.

The following three matrices in Fig. 11 contain the same information for the SN UDA. In general, it can be seen that the proposed model has a much cleaner diagonal, which means fewer confused classes than its competitors. Specifically, there are not as many classes labeled as "herbaceous vegetation" (10) by the models. However, it is still the class with a higher level than other classes associated with it. In the U-Net (C) and DAFormer predictions, the "vineyard" (9) class is highly confused with "agricultural land" (11), "brushwood" (8), or "herbaceous vegetation" (10), with very low accuracy. Nevertheless, even though the proposed model confuses part of the "vineyard" (9) samples with "agricultural land" (11), it is able to obtain much better results.

C. Model Analysis

In Table VIII, an ablation study for the different proposed modules of the DAM-Former is included, in both NS and SN

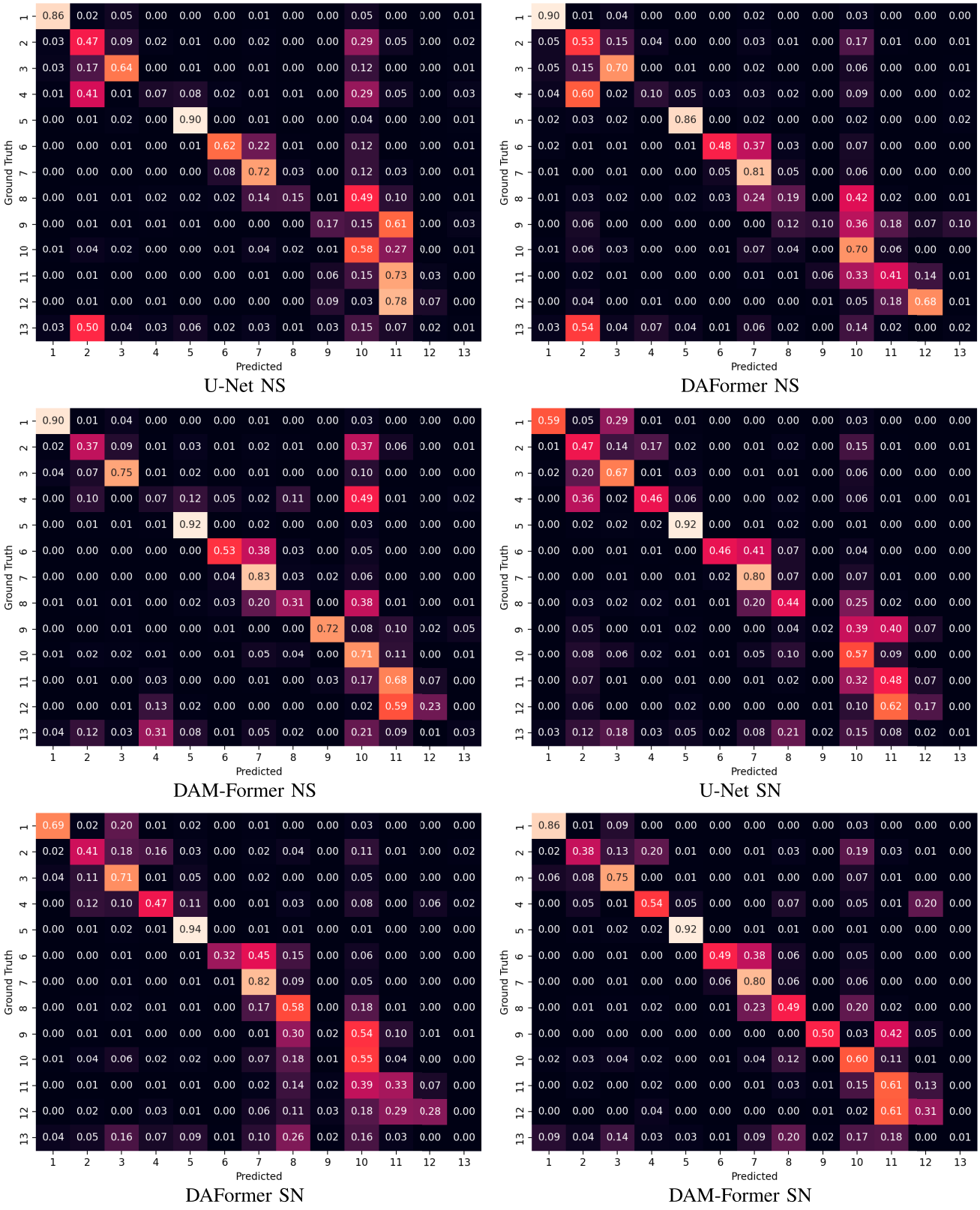


Fig. 11. Normalized confusion matrices of different models using RGB, NIR, elevation, and temporal information for the NS and SN UDA. The numbered semantic classes can be seen in Table I.

scenarios. First, we include the results only using ML. The other modules, the DTF module and the EPs, are added next. In the results, it can be seen that the most relevant module for the performance of the model is the DTF. Indeed, for the NS case, it has the second-best results, and in the SN case, it even reached the best results, obtaining the same mIoU as the final

TABLE XI
QUANTITATIVE ORACLE ANALYSIS IN TERMS OF ACC AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET SN

Class	Time-Series U-Net				Time-Series DAFormer				Ours			
	UDA		ORACLE		UDA		ORACLE		UDA		ORACLE	
N → S	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	76.33	85.57	73.91	80.96	72.36	90.35	72.61	86.95	80.05	90.40	82.89	91.00
P.Surface	27.65	47.29	35.03	41.74	26.91	53.01	37.83	63.03	28.77	37.36	49.45	59.91
I.Surface	55.50	64.06	60.05	71.20	57.82	69.67	60.48	75.38	65.74	74.79	72.66	82.36
B.Soil	5.75	6.87	63.80	82.54	8.29	9.99	67.29	95.85	4.29	7.37	72.39	86.19
Water	71.55	90.03	83.87	96.27	71.38	86.38	83-94	91.86	63.39	91.89	90.56	97.00
Coniferous	31.93	61.92	58.99	74.35	28.38	48.40	39.96	70.97	31.58	53.00	60.73	80.09
Deciduous	62.82	72.38	72.29	80.30	64.07	80.75	66.48	82.45	68.21	82.60	77.22	85.52
Brushwood	13.08	15.07	38.26	51.77	13.80	19.41	36.67	46.16	24.63	21.42	48.63	65.88
Vineyard	12.77	16.76	78.62	84.00	8.80	10.41	72.83	94.42	61.30	71.54	88.69	97.23
H.Vegetation	33.71	57.63	47.15	73.76	40.34	70.08	48.98	75.09	43.33	70.97	56.66	74.58
A.Land	37.66	73.36	53.73	69.70	33.32	40.95	25.60	27.02	47.26	67.63	60.06	70.45
P.Land	6.49	7.21	56.29	77.19	47.41	68.30	48.13	5.29	18.54	22.63	58.97	86.79
Others	0.57	1.47	71.32	85.80	0.71	2.02	23.03	60.50	1.27	2.91	74.33	88.78
AAcc	55.59		72.67		58.78		68.06		64.75		79.26	
MAcc	46.12		74.58		49.98		71.15		54.19		81.98	
mIoU	33.52		61.02		36.43		52.60		41.42		68.71	

TABLE XII
QUANTITATIVE ORACLE ANALYSIS IN TERMS OF ACC AND IOU, AS AACC, MACC, AND MIOU ON THE FLAIR-2 DATASET NS

Class	Time-Series U-Net				Time-Series DAFormer				Ours			
	UDA		ORACLE		UDA		ORACLE		UDA		ORACLE	
S → N	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc	IoU	Acc
Building	56.01	59.40	77.00	85.57	61.37	69.27	77.33	89.43	72.53	86.27	80.68	90.52
P.Surface	26.97	46.54	50.10	66.07	30.98	40.88	42.42	56.06	30.39	38.12	62.72	74.20
I.Surface	48.43	66.64	62.45	74.44	52.25	71.26	64.75	79.65	61.35	75.02	72.45	82.14
B.Soil	10.63	45.82	71.34	95.99	11.28	47.12	68.73	82.98	11.24	54.22	36.17	96.46
Water	70.94	91.82	85.49	88.07	67.38	93.51	76.95	96.55	84.98	91.93	90.31	93.67
Coniferous	36.25	45.53	48.48	72.41	28.51	31.64	43.34	49.44	33.69	49.27	59.86	82.42
Deciduous	64.74	79.80	70.99	83.73	61.33	81.78	69.69	85.77	64.86	79.90	76.58	86.26
Brushwood	25.86	44.45	40.11	51.23	22.12	58.30	37.24	53.57	27.07	48.59	48.76	59.60
Vineyard	1.55	1.73	65.80	68.36	1.20	1.70	75.67	88.82	40.62	49.76	83.97	98.64
H.Vegetation	36.70	57.31	49.20	77.18	35.00	55.46	47.88	64.37	42.96	59.92	58.55	77.48
A.Land	30.81	48.29	41.84	53.25	26.73	33.39	47.03	56.35	37.93	61.07	57.20	67.41
P.Land	14.66	16.96	44.22	52.74	24.10	27.52	54.57	84.99	23.52	30.66	61.51	77.14
Others	0.88	1.08	33.99	69.57	0.21	0.30	74.58	84.82	0.54	0.63	49.56	78.86
AAcc	56.39		71.59		56.66		72.53		63.23		79.29	
MAcc	46.57		72.20		47.09		74.83		55.80		81.91	
mIoU	32.65		57.00		32.50		60.01		40.90		64.49	

model with the EP. These results show that the inclusion of the EPs alone does not have a particular benefit compared to the only ML model. However, once used together with the DTF module, the performance in the NS DA was boosted. On the other hand, in the SN DA, the results were very similar to the best-performing model, showing that it does not generate a negative effect in the event that the model is unable to achieve a better performance. The inclusion of temporal features in the main encoder through the DTF allows the model to enrich its temporal knowledge and better generalize, while the EP module is able to further enhance these capabilities, if possible.

We also compare the fusion approach between the HR multimodal network and the temporal multispectral network. In Table IX, we compare the segmentation results for fusing the features from both networks in all the multiple encoder levels, as proposed in [42], and fusing only the deepest features, as we propose in this work. The proposed DTF module is able to obtain between 5% and 7% better performance in all the metrics. Even though for convolutional encoders, fusing all the multilevel features appears to work, as could also be seen in all the U-Net (C) temporal models included in Section V-B, it could also be seen that for the U-Net (T) transformer-based architecture, the model did not benefit. However, mixing both networks' features only in the deepest encoder increased the performance. This could be due to of the difference in spatial resolution of the shallower features and the information extracted from both networks.

In Table X, we can see the number of parameters of each of the different models from previous experiments. The difference between models using only RGB and the models using RGB + NIR + E data is negligible. The models using exclusively the temporal information have the least number of parameters, with the U-TAE model being the most complex among them. However, they only process the smaller multi-temporal images. Among the models that process aerial images, the U-Net (C) has the lowest number of parameters, being less than half of the U-Net (T) and around 1:4 of the parameters of the DAFormer. The number of parameters

increases when the temporal branch is included in all the cases. Our model has almost the same number of parameters as the transformer-based U-Net, much less than the DAFormer, while obtaining better segmentation results than both of them in the previous experiments. DAM-Former sits in between the highly complex DAFormer and the simpler U-Net while generating the best predictions over the FLAIR-2 dataset.

After analyzing the quantitative and qualitative results, it can be observed that in cases where a large amount of information is available, such as HR images, transformer-based encoders are able to obtain better results than convolutional networks. However, including more information into the system does not always translate into better performance. The fusion and integration of different modalities in UDA need to be performed carefully and may change compared to other tasks with a lower level of complexity. In the experiments, it can be seen that some models even perform worse after incorporating temporal information.

Even though DAM-Former is able to outperform state-of-the-art models in the proposed multimodal UDA task, the predictions still suffer considerably for the case of under-represented and unbalanced classes. Differences among soils, vegetation, and urban structures also change between domains affecting the model.

D. Oracle Analysis

In general, unsupervised models have lower performance than supervised or semi-supervised models. In UDA, when transferring the learned information from a source domain to a different unlabeled target domain, the same trend can be seen. In the following experiment, we compare the results obtained previously in the multimodal models with the same model trained with the same target and source domains (Oracle). The results for the North and South domain adaptation as targets are shown in Tables XI and XII, respectively. In the global metrics, it can be observed how the proposed DAM-Former is able to obtain much higher accuracy results. This means that for a standard image segmentation task, our model would also outperform the competitors. In the tables, it can also be seen that the metrics with very low performance in most of the models obtain a decent Acc and IoU result, such as "bare soil," "brushwood," "vineyard," "plowed land," or "others." The difference in these results points out the challenging UDA between the North and South domains and also shows that these classes are especially different between domains, and it is not only a class balance problem. In contrast, classes such as "building," "impervious surface," or "water" among others are not as heavily affected. Therefore, even though unbalanced classes in data between domains have a considerable impact on UDA, there are other class qualities that can affect the performance of segmentation.

VI. Conclusion

The fusion and use of HR and temporal information in remote sensing for UDA tasks have not yet been explored. This work introduces a new multimodal and multi-temporal UDA model, named DAM-Former, which was designed in the context of remote sensing image segmentation. DAM-Former is composed of two branches: an HR multimodal network with a transformer-based encoder and a lighter convolution-based decoder, to extract fine features; and a lower resolution temporal network, in charge of finding coarse and temporal-dependent information. The two networks are interconnected, sharing information at deep encoding levels. During training, the information from both branches is further fused using a new ML. Furthermore, the final pseudo-labels generated by the model are integrated from both networks, reducing the noise and improving the data fusion.

To analyze the performance of the model, extensive experiments have been carried out with data from the FLAIR-2 dataset, extracting from the North and South domains of this dataset. The results show the performance improvement obtained compared to other architectures and strategies. Future research will focus on tackling the effects of class unbalancing, underrepresentation, and confusing annotated labels. At the same time, we will investigate boosting efficiency during training, enhancing and generalizing multimodal data fusion techniques for other kinds of remote sensing data, such as SAR, LiDAR, or even hyperspectral imagery. Potentially, we could apply our framework to other land cover-related tasks, such as change detection, object detection, or even disaster response.

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